

An aerial photograph of the New York City skyline, featuring the Chrysler Building and the Queensboro Bridge. The image is overlaid with a semi-transparent blue filter. A dark blue rounded rectangle with a white border is centered in the middle of the image, containing the title text in white.

Methods for Contract Submission

Contract Submissions

- There are several ways for agency personnel to submit contract actions to the Comptroller for registration.*
 1. Physical Hand-Delivery
 2. Direct Interface Platforms (PASSPort, Contract Submission Portal, etc.)
 3. Digital File Drop (link required)
- ***Note:** Agencies may also need to file packages associated with self-registered contracts (e.g. Accelerated Procurements and M/WBE Small Purchase Contracts).



Physical Hand-Delivery

Hand-delivered submissions can be made to the Central Imaging Facility (CIF):

- **Address:** 1 Centre, Street, Room No. 1329, New York, NY 10007
 - 13th Floor
- **Phone:** (212) 669-3741

CIF Hours: Monday-Friday, 9am-5pm (closed on City holidays).



Direct Interface Submissions



- **Hours:** 9am-5pm (for same day processing)
- **Reminders:**
 - Direct Interface submissions after these hours will not be considered received until the following business day for the purposes of the Comptroller's 30-day review.
 - Direct Interface made via PASSPort or a similar system will be validated to confirm that approved procurement methods were selected, and that documents corresponding to mandatory fields were attached.
 - If your submission is rejected due to a validation issue, follow error messages guidance or contact the Comptroller's Bureau of Information Systems and Technology (BIST) at helpdesk@comptroller.nyc.gov.



Alternative Digital File Drop Submissions

When agencies can't submit a contract through an electronic system like PASSPort, or via hard copy, Citrix Sharefile and Microsoft Sharepoint allow them to transmit a contract via a single PDF file. Agency staff use these platforms to deposit a contract file in a shared-folder accessible by the Comptroller's office.

- **Requirements:**
 - Email ocamailbox@comptroller.nyc.gov and ClFrequeest@comptroller.nyc.gov to request a link
 - Ensure the Contract #, OCA #, EPIN, and Vendor information are consistent across all contract documents. These values must match what is in FMS.
 - Contracts can only be submitted as a single PDF



Contract Submission Portal (CSP)

- The CSP is a Comptroller-operated contract submission application (housed in Citrix).
- Unlike with Sharefile, agency staff can use the CSP to label individual contract documents.
- Submissions are automatically routed to BCA.
- Interested in signing up?
Email ocamailbox@comptroller.nyc.gov

The screenshot displays the OAISIS Contract Submission Portal interface. At the top, there is a header bar with the title "OAISIS Contract Submission Portal" and a search bar. Below the header, a navigation bar contains several tabs: "Transaction Code", "Contract Number", "Contract Number", "Vendor Number", and "Contract Number". The main content area is organized into a grid of submission fields. Each field has a title, a file upload button, and a status indicator. The fields include: "Advice of Award" (with files "aoa.pdf" and "AOA UPDATED.pdf"), "Agency Memo" (with file "agm.pdf"), "Agreement" (with file "lease.pdf"), "Bid Tabulation", "Budget Detail" (with file "agmemo.pdf"), "Contract Cover Sheet" (with file "cics.pdf"), "DOI Vendor Name Check Memo", "Emergency Declaration", "Insurance Certificate", "Late Registration Memo", "Miscellaneous", "OMB Approval", "Pre-Solicitation Review Report", "Proposal/Bid", "Recommendation For Award", "Recommendation for Renewal", "Responsibility Determination" (with file "w9.pdf"), and "RFP/IFB". Each field also has a "Choose File" button and a "No file chosen" status.



An aerial photograph of the New York City skyline, featuring the Chrysler Building and the Queensboro Bridge. The image is overlaid with a semi-transparent blue filter. A dark blue rounded rectangle with a thin white border is centered in the middle of the image, containing the text "Important Last-Minute Checks" in white.

Important Last-Minute Checks

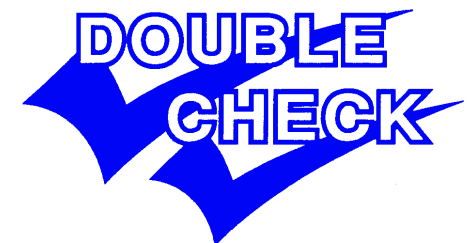
Mandatory Documents

- Each Contract Package must include the following documents:
 - Completed CIF Cover Sheet
 - Final Advice of Award
 - All other appropriate and required documentation pursuant to PPB Section 2-12, or other applicable guidelines.
- For modifications, use the New Document Name Field or the Modification Tab in FMS to enter details about the change.
- Please reach out to the Bureau of Contract Administration if you have any questions about required documentation.



Final Checks Before Submission

- Double check the following before submitting your contract to avoid unnecessary registration delays:
 - Ensure that award method codes, contract type codes, funding codes, and contract terms are correct.
 - Contract/Version #s, OCA #s, Vendor Name/Ids, EPIN #s, and insurance information should be consistent across the Cover Sheet, the Advice of Award, and other contract documents.
 - **Direct Interface Platforms:** Attachments should be properly labeled and filed under the correct document categories.



An aerial photograph of the New York City skyline, featuring prominent skyscrapers like the Chrysler Building and the Empire State Building. The image is overlaid with a semi-transparent blue filter. A dark blue rounded rectangle with a white border is centered in the lower half of the image, containing the text 'FMS Codes' in white.

FMS Codes

4 SCOPE OF WORK

4-1.05 CHANGES AND EXTRA WORK.

4-1.05A Changes Requested by the CONTRACTOR.

4-1.05A(i) General. Changes in the Plans and Specifications, requested in writing by the CONTRACTOR, which do not materially affect the Work and which are not detrimental to the Work or to the interests of the AGENCY, may be granted by the Engineer. Nothing herein shall be construed as granting a right to the CONTRACTOR to demand acceptance of such changes.

4-1.05A(ii) Payment for Changes Requested by the CONTRACTOR. If such changes are granted, they shall be made at a reduction in cost or no additional cost to the AGENCY.

4-1.05B Changes Initiated by the AGENCY.

4-1.05B(i) General. The AGENCY may change the Plans, Specifications, character of the work, or quantity of work provided the total arithmetic dollar value of all such changes, both additive and deductive, does not exceed 25 percent of the Contract Price. Should it become necessary to exceed this limitation, the change shall be by written Supplemental Agreement between the CONTRACTOR and AGENCY, unless both parties agree to proceed with the change by Change Order.

Change Orders shall be in writing and state the dollar value of the change or establish the method of payment, include any adjustment in the Contract time of completion, and when negotiated prices are involved, shall provide for the CONTRACTOR's signature indicating acceptance.

The AGENCY may request that CONTRACTOR provide AGENCY with estimated costs for proposed changes to the work. CONTRACTOR agrees to promptly provide AGENCY with detailed, itemized costs for proposed changes to the work and scheduling data demonstrating the impact, if any, of the proposed changes to the work on the Contract Time. Adjustments, if any, in the amount to be paid the CONTRACTOR by reason of any modifications of the work as set forth in a Contract Change Order, Construction Change Directive, or arising from Claims shall be determined by one or more of the following methods as elected by the AGENCY:

A. Lump Sum Price - By an acceptable lump proposal from the CONTRACTOR. The Lump Sum Price provided by CONTRACTOR shall not include costs or expenses greater than the costs or expenses permitted for Force Account work, as set forth below.

B. Unit Prices - By unit prices fixed by agreement between the AGENCY and the CONTRACTOR under 4-1.05B(ii) below.

C. Force Account – By ordering the CONTRACTOR to proceed with the work and to keep and present in such form as the Engineer or AGENCY may direct, a correct account of the cost of the change, together with all vouchers therefor. The CONTRACTOR will be paid for labor, materials, and equipment rental actually used on the Change Order work as described in 4-1.05C(ii).

4-1.05B(ii) Contract Unit Prices.

(a) General. If a change is ordered in an item of work covered by a Contract Unit Price, and such change does not involve substantial change in character of the work from that shown on the Plans or specified in the Specifications, then an adjustment in payment will be made. This adjustment will be based upon the increase or decrease in quantity and the Contract Unit Price.

If the actual quantity of an item of work covered by Contract Unit Price and constructed in conformance with the Plans and Specifications varies from the Bid quantity by 25 percent or less, payment will be made at the Contract Unit Price. If the actual quantity of said item of work varies from the Bid quantity by more than 25 percent, payment will be made per (b) or (c) below, as appropriate.

If a change is ordered in an item of work covered by a Contract Unit Price, and such change does involve a substantial change in the character of the work from that shown on the Plans or specified in the Specifications, an adjustment in payment will be made per 4-1.05B(iv).

(b) Increases of More than 25 Percent. Should the actual quantity of an item of work covered by a Contract Unit Price and constructed in conformance with the Plans and Specifications, exceed the Bid quantity by more than 25 percent, payment for the quantity in excess of 125 percent of the Bid quantity will be made on the basis of an adjustment in the Contract Unit Price mutually agreed to by the CONTRACTOR and the AGENCY, or at the option of the Engineer, on the basis of Extra Work per 4-1.05C.

The Extra Work per 4-1.05C shall not include fixed costs. Fixed costs shall be deemed to have been recovered by the CONTRACTOR through payment for 125 percent of the Bid quantity at the Contract Unit Price.

(c) Decreases of More Than 25 Percent. Should the actual quantity of an item of work covered by a Contract Unit Price, and constructed in conformance with the Plans and Specifications, be less than 75 percent of the Bid quantity, an adjustment in payment will not be made unless so requested in writing by the CONTRACTOR. If the CONTRACTOR so requests, payment will be made on the basis of an adjustment in the Contract Unit Price mutually agreed to by the CONTRACTOR and the AGENCY, or at the option of the Engineer, on the basis of Extra Work per 4-1.05C; however, in no case will payment be less than would be made for the actual quantity at the Contract Unit Price nor more than would be made for 75 percent of the Bid quantity at the Contract Unit Price.

4-1.05B(iii) Stipulated Unit Prices. Stipulated Unit Prices are unit prices established by the AGENCY in the Contract Documents. Stipulated Unit Prices may be used for the adjustment of Contract changes when so specified in the Special Provisions.

4-1.05B(iv) Agreed Prices. Agreed Prices are prices for new or unforeseen work, or adjustments in Contract Unit Prices per 4-1.05B(ii), established by mutual agreement between the CONTRACTOR and the AGENCY. If mutual agreement cannot be reached, the Engineer may direct the CONTRACTOR to proceed on the basis of Extra Work in accordance 4-1.05C, except as otherwise specified in 4-1.05B(ii)(b) and 4-1.05B(ii)(c).

4-1.05B(v) Eliminated Items. Should any Bid item be eliminated in its entirety, payment will be made to the CONTRACTOR for its actual costs incurred in connection with the eliminated item prior to notification in writing from the Engineer so stating its elimination.

If material conforming to the Plans and Specifications is ordered by the CONTRACTOR for use in the eliminated item prior to the date of notification of elimination by the Engineer, and if the order for the material cannot be canceled, payment will be made to the CONTRACTOR for the actual cost of the materials. In this case, the material shall become the property of the AGENCY. Payment will be made to the CONTRACTOR for its actual costs for any further handling. If the material is returnable, the materials shall be returned and payment will be made to the

CONTRACTOR for the actual cost of charges made by the supplier for returning the material and for handling by the CONTRACTOR.

Actual costs, as used herein, shall be computed on the basis of Extra Work 4-1.05C(ii).

4-1.05C Extra Work.

New or unforeseen work will be classified as “extra work” when the Engineer determines that it is not covered by the Contract Unit Prices, stipulated unit prices, or included in a Lump Sum item.

4-1.05C(i) Payment.

When the price for the extra work cannot be agreed upon, the AGENCY will pay for the extra work based on the accumulation of costs as provided herein.

4-1.05C(ii) Basis for Establishing Costs.

(a) Labor. The costs of labor will be the actual costs for wages of workers performing the extra work at the time the extra work is done, plus employer payments of payroll taxes, workers compensation insurance, liability insurance, health and welfare, pension, vacation, apprenticeship funds, and other direct costs, resulting from Federal, State, or local laws, as well as assessments or benefits required by lawful collection bargaining agreements.

The use of a labor classification which would increase the extra work cost will not be permitted unless the CONTRACTOR establishes the necessity for such additional costs. Labor costs for equipment operators and helpers shall be reported only when such costs are not included in the invoice for equipment rental. The labor costs for foremen shall be proportioned to all of their assigned work and only that applicable to extra work will be paid.

Nondirect labor costs, including superintendence, shall be considered part of the markup of 4-1.05C(iii).

(b) Materials. The cost of materials reported shall be at invoice or lowest current price at which such materials are locally available and delivered to the job site in the quantities involved, plus sales tax, freight, and delivery.

The AGENCY reserves the right to approve materials and sources of supply, or to supply materials to the CONTRACTOR if necessary for the progress of the Work. No markup shall be applied to any materials provided by the AGENCY.

(c) Tool and Equipment Rental. No payment will be made for the use of tools which have a replacement value of \$200 or less.

Regardless of ownership, the rates to be used in determining equipment rental costs shall not exceed listed rates prevailing locally at equipment rental agencies, or distributors, at the time the work is performed.

The rental rates paid shall include the cost of fuel, oil, lubrication, supplies, small tools, necessary attachments, repairs and maintenance of any kind, depreciation, storage, insurance, and all incidentals. Necessary loading and transportation costs for equipment used on the extra work shall be included.

If equipment is used intermittently and, when not in use, could be returned to its rental source at less expense to the AGENCY than holding it at the Work site, it shall be returned, unless the CONTRACTOR elects to keep it at the Work site, at no expense to the AGENCY.

All equipment shall be acceptable to the Engineer, in good working condition, and suitable for the purpose for which it is to be used. Manufacturer's ratings and approved modifications shall be used to classify equipment and it shall be powered by a unit of at least the minimum rating recommended by the manufacturer.

The reported rental time for equipment already at the Work site shall be the duration of its use on the extra work. This time begins when equipment is first put into actual operation on the extra work, plus the time required to move it from its previous site and back, or to a closer site.

(d) Other Items. The AGENCY may authorize other items which may be required on the extra work, including labor, services, materials, and equipment. These items must be different in their nature from those required for the Work, and be of a type not ordinarily available from the CONTRACTOR or Subcontractors.

Invoices covering all such items in detail shall be submitted with the request for payment.

(e) Invoices. Vendors' invoices for materials, equipment rental and other expenditures shall be submitted with the request for payment. If the request for payment is not substantiated by invoices or other documentation, the AGENCY may establish the cost of the item involved at the lowest price which was current at the time of the report.

4-1.05C(iii) Markup.

(a) Work by CONTRACTOR. Unless otherwise provided in the Special Provisions, a reasonable allowance for overhead and profit shall be added to the CONTRACTOR's costs as determined under 4-1.05C(ii) and shall constitute the markup for all overhead and profit on work by the CONTRACTOR. The CONTRACTOR shall also be compensated for the actual increase in the CONTRACTOR's bond premium caused by the extra work.

(b) Work by Subcontractor. When any of the extra work is performed by a Subcontractor, the markup established in (a) above shall be applied to the Subcontractor's costs as determined under 4-1.05C(ii). Unless otherwise provided in the Special Provisions, a reasonable allowance for the CONTRACTOR's overhead and profit shall be added to the sum of the Subcontractor's costs and markup and shall constitute the markup for all overhead and profit for the CONTRACTOR on work by the Subcontractor.

4-1.05C(iv) Daily Reports by CONTRACTOR. When the price for the extra work cannot be agreed upon, the CONTRACTOR shall submit a daily report to the Engineer on forms approved by the AGENCY. Included are applicable delivery tickets, listing all labor, materials, and equipment involved for that day, and other services and expenditures when authorized. Failure to submit the daily report by the close of the next working day may waive any rights for that day. An attempt shall be made to reconcile the report daily, and it shall be signed by the Engineer and the CONTRACTOR. In the event of disagreement, pertinent notes shall be entered by each party to explain points which cannot be resolved immediately. Each party shall retain a signed copy of the report. Reports by Subcontractors or others shall be submitted through the CONTRACTOR.

The report shall:

1. Show names of workers, classifications, and hours worked.
2. Describe and list quantities of materials used.

3. Show type of equipment, size, identification number, and hours of operations, including loading and transportation, if applicable.

4. Describe other services and expenditures in such detail as the AGENCY may require.

4-1.05D Changed Conditions. The CONTRACTOR shall promptly notify the Engineer of the following Work site conditions (hereinafter called changed conditions), in writing, upon their discovery and before they are disturbed.

1. Subsurface or latent physical conditions differing materially from those represented in the Contract;

2. Unknown physical conditions of an unusual nature differing materially from those ordinarily encountered and generally recognized as inherent in work of the character being performed; and

3. Material differing from that represented in the Contract which the CONTRACTOR believes may be hazardous waste, as defined in section 25117 of the Health and Safety Code, that is required to be removed to a Class I, Class II, or Class III disposal site in accordance with provisions of existing law.

The Engineer will promptly investigate conditions which appear to be changed conditions. If the Engineer determines that the conditions are changed conditions and will materially affect costs, a Change Order will be issued adjusting the compensation for such portion of the Work in accordance with 4-1.05B(ii). If the Engineer determines that conditions are changed conditions and they will materially affect performance time, the CONTRACTOR, upon submitting a written request, will be granted an extension of time subject to the provisions of 8-1.08.

If the Engineer determines that the conditions do not justify an adjustment in compensation, the CONTRACTOR will be notified in writing. This notice will also advise the CONTRACTOR of its obligation to notify the Engineer in writing if the CONTRACTOR disagrees.

Should the CONTRACTOR disagree with the decision, it may submit a written notice of potential claim to the Engineer before commencing the disputed work. In the event of such a dispute, the CONTRACTOR shall not be excused from any scheduled completion date provided by the Contract and shall proceed with all work to be performed under the Contract. However, the CONTRACTOR shall retain any and all rights provided by either Contract or law which pertain to the resolution of disputes and protests between the contracting parties. The CONTRACTOR shall proceed as provided in 4-1.05E.

The CONTRACTOR's failure to give notice of changed conditions promptly upon their discovery and before they are disturbed shall constitute a waiver of all claims in connection therewith.

4-1.05E Disputed Work. If the CONTRACTOR and the AGENCY are unable to reach agreement on disputed work, the AGENCY may direct the CONTRACTOR to proceed with the work. Payment shall be as later determined by mediation or arbitration, if the AGENCY and CONTRACTOR agree thereto, or as fixed in a court of law.

Although not to be construed as proceeding under extra work provisions, the CONTRACTOR shall keep and furnish records of disputed work in accordance with 4-1.05C.

5 CONTROL OF WORK

5-1.02 CONTRACT COMPONENTS.

5-1.02A General.

The CONTRACTOR shall ascertain the existence of any conditions affecting the cost of the Work through a reasonable examination of the Work site prior to submitting the Bid.

Existing improvements visible at the Work site, for which no specific disposition is made on the Plans, but which interfere with the completion of the Work, shall be removed and disposed of by the CONTRACTOR.

The CONTRACTOR shall, upon discovering any error or omission in the Plans or Specifications, immediately call it to the attention of the Engineer.

All final locations determined in the field, and any deviations from the Plans and Specification, shall be marked in red on the documents to show the as-built conditions. CONTRACTOR shall maintain a complete and accurate record of all changes of construction from that shown in these plans and specifications for the purpose of providing a basis for construction record drawings. No changes shall be made without prior written approval of the Engineer. Upon completion of the Project, CONTRACTOR shall deliver this record of all construction changes to the Engineer along with a letter which declares that other than these noted changes “the Project was constructed in conformance with the Contract Documents.” Final payment will not be made until this requirement is met.

As the figured dimensions shown on the drawings and in the specifications of the Contract may not in every case agree with scaled dimensions, the figured dimensions shall be followed in preference to the scaled dimensions, and drawings to a large scale shall be followed in preference to the drawings to a small scale. Should it appear that the work to be performed, or any related matter, are not sufficiently detailed or explained in the Contract documents, the CONTRACTOR shall apply to the Engineer for such further explanations as necessary, and shall conform to such further explanations provided by the Engineer as part of the Contract to the extent that it is consistent with the terms of the Contract.

Caution: The engineer preparing these plans will not be responsible or liable for unauthorized changes to or uses of these plans. All changes to the plans must be approved in writing by the Engineer.

5-1.02B Records of Construction Changes / As-Built. CONTRACTOR shall maintain a complete and accurate record of all changes of construction from that shown in these Plans and Specifications for the purpose of providing a basis for construction record drawings. No changes shall be made without prior written approval of the City Engineer.

Upon completion of the Project, CONTRACTOR shall deliver this record of all construction changes to the Engineer along with a letter which declares that other than these noted changes that Project was constructed in conformance with the Contract Documents.

Caution: The engineer preparing these Plans will not be responsible for, or liable for, unauthorized changes to or uses of these Plans. All changes to the Plans must be approved in writing by City Engineer.

5-1.02C Precedence of Contract Documents.

If there is a conflict between any of the Contract Documents, the document highest in precedence shall control. The order of precedence, from highest to lowest, shall be as follows:

- 1) Permits issued by jurisdictional regulatory agencies.
- 2) Change Orders and/or Supplemental Agreements; whichever occurs last.
- 3) Contract/Agreement
- 4) Addenda.
- 5) Bid/Proposal.
- 6) Special Provisions.
- 7) Plans.
- 8) Notice Inviting Bids
- 9) Instructions to Bidders
- 10) Standard Plans.
- 11) Standard Specifications.
- 12) Reference Specifications.

5-1.03 ENGINEER'S AUTHORITY. Add *the following*:

The Engineer has the authority to enforce compliance with the Plans and Specifications. The CONTRACTOR shall promptly comply with instructions from the Engineer or an authorized representative.

The decision of the Engineer is final and binding on all questions relating to: quantities; acceptability of materials, equipment, or work; execution, progress or sequence of work; and interpretation of the Plans, Specifications, or other drawings. This shall be precedent to any payment under the Contract, unless otherwise ordered by the Board.

5-1.12 ASSIGNMENT. *Substitute the following*:

No Contract or portion thereof may be assigned without consent of the City Council, except that the CONTRACTOR may assign money due or which will accrue to it under the Contract. If given written notice, such assignment will be recognized by the City Council to the Extent permitted by law. Any assignment of money shall be subject to all property withholdings in favor of the AGENCY and to all deductions provided for in the Contract. All money withheld, whether assigned or not, shall be subject to being used by the AGENCY for completion of the Work, should the CONTRACTOR be in default.

5-1.13 SUBCONTRACTING. *Substitute the following*:

5-1.13A General. Each Bidder shall comply with the Public Contract Code including sections 4100 through 4113. The following excerpts or summaries of some of the requirements of this Chapter are included below for information:

The Bidder shall set for the in the Bid, as provided in 4104:

“(a) The name, and location of the place of business, the California contractor license number, and public works contractor registration number issued pursuant to section 1725.5 of the Labor Code of each subcontractor who will perform work or labor or render service to the prime CONTRACTOR in or about the construction of the work or improvement, or a subcontractor

licensed by the State of California who, under subcontract to the prime CONTRACTOR, specially fabricates and installs a portion of the work or improvement according to detailed drawings contained in the plans and specifications, in an amount in excess of one-half of 1 percent of the prime CONTRACTOR's total bid, or, in the case of bids or offers for the construction of streets or highways, including bridges, in excess of one-half of 1 percent of the prime CONTRACTOR's total bid or ten thousand dollars (\$10,000), whichever is greater."

"(b) The portion of the work which will be done by each such subcontractor under this act. The prime CONTRACTOR shall list only one subcontractor for each such portion as is defined by the prime CONTRACTOR in his bid."

Subcontracting of more than one-half of one percent of the work for which no Subcontractor was designated in the original Bid will be allowed only in cases of public emergency or necessity and only after the Engineer makes a written finding of circumstances constituting public emergency or necessity.

The CONTRACTOR must obtain written consent of the AGENCY to substitute a Subcontractor designated in the original Bid, to permit any subcontract to be assigned or transferred, or to otherwise allow a subcontract to be performed by anyone other than the originally designated Subcontractor.

A violation of any of the above provisions will be considered a violation of the Contract, and the AGENCY may cancel the Contract and collect appropriate damages or assess the CONTRACTOR a penalty of not more than ten (10) percent of the subcontract involved.

If subcontracted work is not being performed in a satisfactory manner, the AGENCY will notify the CONTRACTOR of the need to take corrective action and the Engineer may report the facts to the City Council. Upon order by City Council and the CONTRACTOR's receipt of written instructions from the Engineer, the Subcontractor shall immediately be removed from the Work and may not again be employed on the Work.

5-1.13B Additional Responsibility. The CONTRACTOR shall perform, with its own organization, Contract work amounting to at least 50 percent of the Contract Price except that any designated "Specialty Items" may be performed by subcontract and the amount of any such "Specialty Items" so performed may be deducted from the Contract Price before computing the amount required to be performed by the CONTRACTOR with its own organization. "Specialty Items" will be identified by the AGENCY in the Bid or Proposal. Where an entire item is subcontracted, the value of work subcontracted will be based on the Contract Unit Price. When a portion of an item is subcontracted, the value of work subcontracted will be based on the estimated percentage of the Contract Unit Price. This will be determined from information submitted by the CONTRACTOR, and subject to approval by the Engineer.

5-1.13C Status of Subcontractors. All persons engaged in the Work, including Subcontractors and their employees, will be considered employees of the CONTRACTOR. The CONTRACTOR will be held responsible for their work. The AGENCY will deal directly and solely with the CONTRACTOR and make all payments to the CONTRACTOR.

5-1.16 REPRESENTATIVE. *Add the following:*

5-1.16A Contractor's Representative. Before starting work, the CONTRACTOR shall designate in writing a representative who shall have complete authority to act for it. An alternative representative may be designated as well. The representative or alternate shall be present at the

Work site whenever work is in progress or whenever actions of the elements necessitate its presence to take measures necessary to protect the Work, persons, or property. Any order or communication given to this representative shall be deemed delivered to the CONTRACTOR. A joint venture shall designate only one representative and alternate. In the absence of the CONTRACTOR or its representative, instructions or directions may be given by the Engineer to the superintendent or person in charge of the specific work to which the order applies. Such order shall be complied with promptly and referred to the CONTRACTOR or its representative.

In order to communicate with the AGENCY, the CONTRACTOR's representative, superintendent, or person in charge of specific work shall be able to speak, read, and write the English language.

5-1.20 COORDINATION WITH OTHER ENTITIES. *Add the following:*

5-1.20B(5) Permits. Prior to the start of any work, the CONTRACTOR shall apply for and receive any applicable City, County, State, and Federal permits.

The CONTRACTOR shall pay all business taxes or license fees that are required for the work.

All costs associated with these permits are responsibility of CONTRACTOR. If applicable, CONTRACTOR is required to obtain a no fee City Encroachment permit for this project and comply with all permit conditions.

5-1.20B(6) Cooperation and Collateral Work. The CONTRACTOR shall be responsible for ascertaining the nature and extent of any simultaneous, collateral, and essential work by others. The AGENCY, its workers and CONTRACTORS and others, shall have the right to operate within or adjacent to the Work site during the performance of such work.

The AGENCY, the CONTRACTOR, and each of such workers, CONTRACTORS and others, shall coordinate their operations and cooperate to minimize interference.

The CONTRACTOR shall include in its Bid all costs involved as a result of coordinating its work with others. The CONTRACTOR will not be entitled to additional compensation from the AGENCY for damages resulting from such simultaneous, collateral, and essential work. If necessary to avoid or minimize such damage or delay, the CONTRACTOR shall redeploy its work force to other parts of the Work.

Should the CONTRACTOR be delayed by the AGENCY, and such delay could not have been reasonably foreseen or prevented by the CONTRACTOR, the Engineer will determine the extent of the delay, the effect on this project, and any extension of time.

CONTRACTOR shall coordinate his/her work to minimize disruption to ongoing or scheduled private development projects in the project area.

5-1.23 SUBMITTALS. *Substitute the following:*

5-1.23A General. Submittals shall be provided, at the CONTRACTOR's expense, as required by the Plans or Special Provisions, or when requested by the Engineer.

Materials shall neither be furnished nor fabricated, nor shall any work for which submittals are required be performed, before the required submittals have been reviewed and accepted by the Engineer. Neither review nor acceptance of submittals by the Engineer shall relieve the CONTRACTOR from responsibility for errors, omissions, or deviations from the Contract Documents, unless such deviations were specifically called to the attention of the Engineer in the

letter of transmittal. The CONTRACTOR shall be responsible for the correctness of the submittals.

The CONTRACTOR shall allow a minimum of 20 working days for review of submittals unless otherwise specified in the Special Provisions. Each submittal shall be accompanied by a letter of transmittal.

5-1.23B Working Drawings. Working drawings are drawings showing details not shown on the Plans which are required to be designed by the CONTRACTOR. Working drawings shall be of a size and scale to clearly show all necessary details.

Six copies and one reproducible shall be submitted. If no revisions are required, three of the copies will be returned to the CONTRACTOR. If revisions are required, the Engineer will return one copy along with the reproducible for resubmission. Upon acceptance, the Engineer will return two of the copies to the CONTRACTOR and retain the remaining copies and the reproducible.

5-1.23C Shop Drawings. Shop drawings are drawings showing details of manufactured or assembled products proposed to be incorporated into the Work.

5-1.23C Supporting Information. Supporting information is information required by the Specifications for the purposes of administration of the Contract, analysis for verifications of conformance with the Specifications, the operation and maintenance of a manufactured product or system to be constructed as part of the Work, and other information as may be required by the Engineer. Supporting information for systems shall be bound together and include all manufactured items for the system. If resubmittal is not required, three copies will be returned to the CONTRACTOR. Supporting information shall consist of the following and is required unless otherwise specified in the Special Provisions:

- 1) List of Subcontractors.
- 2) List of Materials.
- 3) Certifications.
- 4) Construction Schedule.
- 5) Confined Space Entry Program.
- 6) Concrete mix designs.
- 7) Asphalt concrete mix designs.
- 8) Data, including, but not limited to, catalog sheets, manufacturer's brochures, technical bulletins, specifications, diagrams, product samples, and other information necessary to describe a system, product or item. This information is required for irrigation systems, street lighting systems, and traffic signals, and may also be required for any product, manufactured item, or system.

5-1.37 MAINTENANCE AND PROTECTION

5-1.37A General.

The CONTRACTOR shall be responsible for the protection of public and private property adjacent to the Work and shall exercise due caution to avoid damage to such property.

The CONTRACTOR shall relocate, repair, replace or reestablish all existing improvements within the project limits (e.g., curbs, sidewalks, catch basins, catch basin screens, driveways, fences, walls, sprinkler systems, signs, utility installations, traffic loops, pavements, structures, survey monuments, landscaping, etc.) that are damaged or removed as a result of the CONTRACTOR's operations or as required by the plans and specifications.

All existing improvements, either within the right-of-way or not, including irrigation lines that are damaged by actions of the CONTRACTOR, shall be restored by the CONTRACTOR to their original or better condition at the CONTRACTOR's expense.

The CONTRACTOR shall mark, as approved by the Engineer, all survey monuments, manholes, valves, substructures, or other items that are visible on the surface and will be covered by his operations. This shall be completed prior to the start of that operation and approved by the Engineer.

Existing traffic striping, pavement markings, centerline reflective markers, and curb markings shall also be considered as existing improvements and the CONTRACTOR shall repaint or replace in kind, at the CONTRACTOR's expense, such striping or markings (except for traffic striping and pavement markings within the limits of the Work) if damaged or if their reflectivity is reduced due to construction operations.

All restoration of existing improvements must occur within the construction completion date, unless directed otherwise by the City Engineer.

Maintenance of street and traffic signal systems that are damaged, temporarily removed or relocated shall be done in conformance with these specifications.

Trees, lawns, and shrubbery that are not to be removed shall be protected from damage or injury. If damaged or removed due to CONTRACTOR's operations, they shall be restored or replaced in as nearly the original condition and location as is reasonably possible. Lawns shall be reseeded and covered with suitable mulch.

The CONTRACTOR shall give reasonable notice to occupants or owners of adjacent property to permit them to salvage or relocate plants, trees, fences, sprinklers, and other improvements, within the right-of-way which are designated for removal and would be destroyed because of the Work.

All costs to the CONTRACTOR for protecting, removing, restoring, relocating, repairing, replacing, or reestablishing existing improvements shall be the responsibility of the CONTRACTOR.

The CONTRACTOR shall also abate dust nuisance by cleaning, sweeping and sprinkling with water, or other means as necessary. The use of water resulting in mud on public streets will not be permitted as a substitute for sweeping or other methods.

Materials and equipment shall be removed from the site as soon as they are no longer necessary. Before the final inspection, the site shall be cleared of equipment, unused materials, and rubbish so as to present a satisfactory clean and neat appearance. All cleanup costs shall be included in the CONTRACTOR's Bid.

Excess excavated material from catch basins or similar structures shall be removed from the site immediately. Sufficient material may remain for use as backfill if permitted by the Specifications. Forms and form lumber shall be removed from the site as soon as practicable after stripping.

5-1.37C PROJECT SITE MAINTENANCE.

5-1.37C(i) Cleanup and Dust Control. Throughout all phases of construction, including suspension of work, and until the final acceptance, the CONTRACTOR shall keep the site clean and free from rubbish and debris. The CONTRACTOR shall also abate dust nuisance by cleaning, sweeping and sprinkling with water, or other means as necessary. The use of water resulting in mud on public streets will not be permitted as a substitute for sweeping or other methods.

When required by the Plans or Specifications, the CONTRACTOR shall furnish and operate a self-loading motor sweeper with spray nozzles at least once each working day for the purpose of keeping paved areas acceptably clean wherever construction, including restoration, is incomplete.

Materials and equipment shall be removed from the site as soon as they are no longer necessary. Before the final inspection, the site shall be cleared of equipment, unused materials, and rubbish so as to present a satisfactory clean and neat appearance. All cleanup costs shall be included in the CONTRACTOR's Bid.

Care shall be taken to prevent spillage on haul routes. Any such spillage shall be removed immediately, and the area cleaned.

Excess excavated material shall be removed from the site immediately. Sufficient material may remain for use as backfill if permitted by the Specifications. Forms and form lumber shall be removed from the site as soon as practicable after stripping.

Failure of the CONTRACTOR to comply with the Engineer's cleanup orders may result in an order to suspend work until the condition is corrected. No additional compensation will be allowed as a result of such suspension.

5-1.37C(i)(a) Work Area Appearance.

The CONTRACTOR shall maintain a neat appearance to the Work at all times.

All unsuitable construction materials and rubbish and debris shall be regularly removed from the job site, be transported to a suitable location, and be disposed of in a proper and legal manner. Materials which are to be disposed of shall not be stored at the project sites but shall be removed before the end of the each working day.

In any area visible to the public, the following shall apply:

1. Broken concrete and debris developed during clearing and grubbing shall be disposed of weekly.
2. The CONTRACTOR shall furnish trash bins for all debris from structure construction. All debris shall be placed in trash bins daily.
3. Forms or false work that are to be re-used shall be neatly stacked concurrent with their removal.
4. Forms and false work that are not to be re-used shall be disposed of with their removal.
5. Wash down from concrete trucks shall be at one location in accordance with SWPPP or ESCP as applicable. Concrete from wash down procedures shall be removed from the site weekly.

5-1.37C(ii) Air Pollution Control. The CONTRACTOR shall not discharge smoke, dust, or any other air contaminants into the atmosphere in such quantity as will violate the regulations of any legally constituted authority.

5-1.37C(iii) Vermin Control. At the time of acceptance, structures entirely constructed under the Contract shall be free of rodents, insects, vermin, and pests. Necessary extermination work shall be arranged and paid for by the CONTRACTOR as part of the Work within the Contract time, and shall be performed by a licensed exterminator in accordance with requirements of governing authorities. The CONTRACTOR shall be liable for injury to persons or property and responsible for the elimination of offensive odors resulting from extermination operations.

5-1.37C(iv) Sanitation. The CONTRACTOR shall provide and maintain enclosed contained toilets for the use of employees engaged in the Work. These accommodations shall be maintained in a neat and sanitary condition. They shall also comply with all applicable laws, ordinances, and regulations pertaining to public health and sanitation of dwellings and camps.

Wastewater shall not be interrupted. Any interruption of wastewater facilities shall be immediately reported to AGENCY. Should the CONTRACTOR disrupt existing sewer facilities, sewage shall be conveyed in closed conduits and disposed of in a sanitary sewer system. Sewage shall not be permitted to flow in trenches or be covered by backfill.

5-1.37C(v) Temporary Light, Power, and Water. The CONTRACTOR shall furnish, install, maintain, and remove all temporary light, power, and water at its own expense. Only non-potable water shall be used for construction activities. These include piping, wiring, lamps, and other equipment necessary for the Work. The CONTRACTOR shall not draw water from any fire hydrant (except to extinguish a fire), without obtaining permission from the water agency concerned.

5-1.37C(vi) Water Pollution Control. The CONTRACTOR shall exercise every reasonable precaution to protect channels, storm drains, and bodies of water from pollution. It shall conduct and schedule operations to minimize or avoid muddying and silting of said channels, drains, and waters. Water pollution control work shall consist of constructing those facilities which may be required to provide prevention, control, and abatement of water pollution.

5-1.37C(vi)(a) General. This item shall consist of preparation, implementation and compliance with a storm water pollution prevention plan (SWPPP) or ESCP for the project, if applicable.

5-1.37C(vi)(b) Storm Water Pollution Prevention Plan (SWPPP) Preparation. CONTRACTOR shall submit to the engineer a completed and signed SWPPP or ESCP at the preconstruction conference. The plan may utilize the practices recommended in the latest edition of the California Storm Water Best Management Practices Handbook, available from California Stormwater Quality Association (CSQA), and online at <http://www.cabmphandbooks.net/>. The plan shall be consistent with the construction General Permit, issued by the State Water Resources, Control City Council, through submittal of the Notice of Intent (NOI).

If construction will occur between October 15 and April 15 (considered as the rainy season per City Ordinance), a wet weather erosion control plan must be submitted. Additionally, Best Management Practices (BMPs) implemented during the AGENCY's rainy season shall include but not be limited to those appropriate for wet weather conditions.

5-1.37C(vi)(c) Storm Water Pollution Prevention Measures. All storm water pollution prevention measures shall be in accordance with the submitted SWPPP or ESCP. In the event circumstances during the course of construction require changes to the original SWPPP or ESCP, a revised plan shall be promptly submitted to the AGENCY's representative in each instance. No responsibility shall accrue to the AGENCY as a result of the plan or as a result of knowledge of

the plan. All work installed by the CONTRACTOR in connection with the SWPPP or ESCP but not specified to become a permanent part of the project shall be removed and the site restored in so far as practical to its original condition prior to completion of construction or when directed by the AGENCY's representative.

5-1.37C(vi)(d) Storm Water Pollution Prevention – Measurement and Payment. Unless otherwise indicated in the Special Provisions, measurement and payment for Storm Water Pollution Prevention Measures, as described herein, shall be included in the items of Work requiring storm water pollution prevention measures as indicated in the project Special Provisions. Such payment shall be considered full compensation for all labor, materials, tools, and equipment for completion, and implementation and compliance with the SWPPP or ESCP.

5-1.37C(vii) Drainage Control. The CONTRACTOR shall maintain drainage within and through the work areas. Earth dams will not be permitted in paved areas. Temporary dams of sandbags, asphaltic concrete, or other acceptable material will be permitted when necessary. Such dams shall be removed from the site as soon as their use is no longer necessary.

5-1.43 POTENTIAL CLAIMS AND DISPUTE RESOLUTION.

5-1.43G Construction Materials Dispute Resolution (Soils, Rock Materials, Concrete, Mortar and Related Materials, Masonry Materials, Bituminous Materials, Rock Products, and Modified Asphalts).

In the interest of safety and public value, whenever credible evidence arises to contradict the test values of materials, the AGENCY and the CONTRACTOR will initiate, an immediate and cooperative investigation. Test values of materials are results of the materials' tests, as defined by these Specifications or by the special provisions, required to accept the Work. Credible evidence is process observations or test values gathered using industry accepted practices. A contradiction exists whenever work acceptance or performance becomes suspect. The investigation shall allow access to all test results, procedures, and facilities relevant to the disputed work and consider all available information and, when necessary, gather new and additional information to determine the validity, the cause, and if necessary, the remedy to the contradiction. If the cooperative investigation reaches any resolution mechanism acceptable to both the AGENCY and the CONTRACTOR, the contradiction shall be considered resolved and the cooperative investigation concluded.

Whenever the cooperative investigation is unable to reach resolution, the investigation may then either conclude without resolution or continue by written notification of one party to the other requesting the implementation of a resolution process by committee. The continuance of the investigation shall be contingent upon recipient's agreement and acknowledged in writing within 3 calendar days after receiving a request. Without acknowledgement, the investigation shall conclude without resolution. The committee shall consist of three State of California Registered Civil Engineers. Within 7 calendar days after the written request notification, the AGENCY and the CONTRACTOR will each select one engineer. Within 14 calendar days of the written request notification, the two selected engineers will select a third engineer. The goal in selection of the third member is to complement the professional experience of the first two engineers. Should the two engineers fail to select the third engineer, the AGENCY and the CONTRACTOR shall each propose 2 engineers to be the third member within 21 calendar days after the written request notification. The first two engineers previously selected shall then select one of the court proposed engineers in a blind draw.

The committee shall be a continuance of the cooperative investigation and will re-consider all available information and if necessary, gather new and additional information to determine the validity, the cause, and if necessary, the remedy to the contradiction. The committee will focus upon the performance adequacy of the material(s) using standard engineering principles and practices and to ensure public value, the committee may provide engineering recommendations as necessary. Unless otherwise agreed, the committee will have 30 calendar days from its formation to complete their review and submit their findings. The final resolution of the committee shall be by majority opinion, in writing, stamped and signed. Should the final resolution not be unanimous, the dissenter may attach a written, stamped, and signed minority opinion.

Once started, the resolution process by committee shall continue to full conclusion unless:

1. Within 7 days of the formation of the committee, the AGENCY and the CONTRACTOR reach an acceptable resolution mechanism; or
2. Within 14 days of the formation of the committee, the initiating party withdraws its written notification and agrees to bear all investigative related costs thus far incurred; or
3. At any point by the mutual agreement of the AGENCY and the CONTRACTOR.

Unless otherwise agreed, the CONTRACTOR shall bear and maintain a record for all the investigative costs until resolution. Should the investigation discover assignable causes for the contradiction, the assignable party, the AGENCY or the CONTRACTOR, shall bear all costs associated with the investigation. Should assignable causes for the contradiction extended to both parties, the investigation will assign costs cooperatively with each party or when necessary, equally. Should the investigation substantiate a contradiction without assignable cause, the investigation will assign costs cooperatively with each party or when necessary, equally. Should the investigation be unable to substantiate a contradiction, the initiator of the investigation shall bear all investigative costs. All claim notification requirements of the contract pertaining to the contradiction shall be suspended until the investigation is concluded.

5-1.46 FINAL INSPECTION AND CONTRACT ACCEPTANCE.

The Work will be inspected by the Engineer for acceptance upon receipt of the CONTRACTOR's written assertion that the Work has been completed.

If, in the Engineer's judgment, the Work has been completed and is ready for acceptance, it will so certify to the City Council, which may accept the completed Work. The Engineer will, in its certification to the City Council, give the date when the Work was completed. This will be the date when the CONTRACTOR is relieved from responsibility to protect the Work.

All work shall be warranted by the CONTRACTOR against defective workmanship and materials for a period of 1 year from the date a Notice of Completion is filed. The CONTRACTOR shall replace or repair any such defective work in a manner satisfactory to the Engineer, after notice to do so from the Engineer, and within the time specified in the notice. If the CONTRACTOR fails to make such replacement or repairs within the time specified in the notice, the AGENCY may perform this work and the CONTRACTOR's sureties shall be liable for the cost thereof.

5-1.47 GUARANTEE.

The CONTRACTOR shall warrant and guarantee the entire Work and all parts thereof, including that performed and constructed by subcontractors, and others employed directly or indirectly on the Work, against faulty or defective materials, equipment or workmanship for the maximum

period provided by law. In addition thereto, for a period of one (1) year commencing on the date of filing of the Notice of Completion, the CONTRACTOR shall, upon the receipt of notice in writing from the AGENCY, promptly make all repairs arising out of defective materials, workmanship or equipment and bear the cost thereof. The AGENCY is hereby authorized to make such repairs and the CONTRACTOR and Surety shall bear the cost thereof if, ten (10) days after the giving of such notice to the CONTRACTOR, the CONTRACTOR has failed to make or undertake with due diligence the repairs; provided, however, that, in the case of an emergency where, in the opinion of the AGENCY, delay could cause serious loss or damage, repairs may be made without notice being sent to the CONTRACTOR or Surety, and all expense in connection therewith shall be charged to the CONTRACTOR and Surety.

For the purpose of this article "Acceptance of the Work" shall mean the acceptance of the Work by the AGENCY in accordance with 8-1.10 but not for the purpose of extinguishing any covenant or agreement or agreement on the part of the CONTRACTOR to be performed or fulfilled under this Contract, which has not in fact been performed or fulfilled at the time of such acceptance all of such covenants and agreements, shall continue to be binding on the CONTRACTOR until they have been fulfilled.

The effective date of Acceptance of the Work and commencement of the Guarantee shall be the date of acceptance of the Notice of Completion by the City Council.

6 CONTROL OF MATERIALS

6-1.01 General.

All materials, parts, and equipment furnished by the CONTRACTOR in the Work shall be new, high grade, and free from defects. Quality of work shall be in accordance with the generally accepted standards. Materials and work quality shall be subject to the Engineer's approval.

Materials and work quality not conforming to the requirements of the Specifications shall be considered defective and will be subject to rejection. Defective work or material, whether in place or not, shall be removed immediately from the site by the CONTRACTOR, at its expense, when so directed by the Engineer.

If the CONTRACTOR fails to replace any defective or damaged work or materials after reasonable notice, the Engineer may cause such work or materials to be replaced. The replacement expense will be deducted from the amount to be paid to the CONTRACTOR.

Used or secondhand materials, parts, and equipment may be used only if permitted by the Specifications.

The CONTRACTOR and all subcontractors, suppliers, and vendors shall guarantee that the entire Work will meet all requirements of this Contract as to the quality of materials, equipment, and workmanship. The CONTRACTOR, at no cost to the AGENCY, shall make any repairs or replacements made necessary by defects in materials, equipment, or workmanship that become evident within one year after the date of recordation of the Notice of Completion. Within this one year period, the CONTRACTOR shall also restore to full compliance with the requirements of this Contract any portion of the Work which is found not to meet those requirements. The CONTRACTOR shall defend, indemnify, and hold the AGENCY, its officers, agents, and employees harmless from claims of any kind due to injuries or damages arising, directly or indirectly, from said defects or noncompliance.

The CONTRACTOR shall make all repairs, replacements, and restorations within thirty-five (35) days after the date of the Engineer's written notice.

If, in the opinion of the Engineer, the defective work is not of sufficient magnitude or importance to make the work dangerous or undesirable, or if, in the opinion of the Engineer, the removal of such work is impractical or will create conditions which are dangerous or undesirable, the AGENCY shall have the right and authority to retain such work instead of requiring it to be removed and reconstructed, but will make such deductions thereof in the payments due or to become due to the CONTRACTOR as the AGENCY may deem just and reasonable.

6-1.01A Protection of Work and Materials. The CONTRACTOR shall provide and maintain storage facilities and employ such measures as will preserve the specified quality and fitness of materials to be used in the Work. Stored materials shall be reasonably accessible for inspection. The CONTRACTOR shall also adequately protect new and existing work and all items of equipment for the duration of the Contract.

The CONTRACTOR shall not, without the AGENCY's consent, assign, sell, mortgage, hypothecate, or remove equipment or materials which have been installed or delivered and which may be necessary for the completion of the Contract.

6-1.01B Inspection Requirements.

6-1.01B(i) General. Unless otherwise specified, inspection is required at the source for such typical materials and fabricated items as bituminous paving mixtures, structural concrete, metal fabrication, metal casting, welding, concrete pipe manufacture, protective coating application, and similar shop or plant operations.

Steel pipe in sizes less than 450 mm (18 inches) and vitrified clay and cast iron pipe in all sizes are acceptable upon certification as to compliance with the specifications, subject to sampling and testing by the AGENCY. Standard items of equipment such as electric motors, conveyors, elevators, plumbing fixtures, etc., are subject to inspection at the job site only. Special items of equipment such as designed electrical panel boards, large pumps, sewage plant equipment, etc., are subject to inspection at the source, normally only for performance testing. The Specifications may require inspection at the source for other items not typical of those listed in this Section.

6-1.01B(ii) Inspection of Materials Not Locally Produced. When the CONTRACTOR intends to purchase materials, fabricated products, or equipment from sources located more than 80 km (50 miles) outside the geographical limits of the AGENCY, an inspector or accredited testing laboratory (approved by the Engineer), shall be engaged by the CONTRACTOR at its expense, to inspect the materials, equipment or process. This approval shall be obtained before producing any material or equipment. The inspector or representative of the testing laboratory shall judge the materials by the requirements of the Plans and Specifications. The CONTRACTOR shall forward reports required by the Engineer. No materials or equipment shall be shipped nor shall any processing, fabrication or treatment of such materials be done without proper inspection by the approved agent. Approval by said agent shall not relieve the CONTRACTOR of responsibility for complying with the Contract requirements.

6-1.01B(iii) Inspection by the AGENCY. The AGENCY will provide all inspection and testing laboratory services within 80 km (50 miles) of the geographical limits of the AGENCY. For private contracts, all costs of inspection at the source, including salaries and mileage costs, shall be paid by the permittee.

6-1.01B(iv) Test of Materials. Before incorporation in the Work, the CONTRACTOR shall submit samples of materials, as the Engineer may require, at no cost to the AGENCY. The CONTRACTOR, at its expense, shall deliver the materials for testing to the place and at the time designated by the Engineer. Except as elsewhere specified, the AGENCY will bear the cost of testing material and/or workmanship which meet or exceed the requirements indicated in the Standard Specifications and the Special Provisions. The CONTRACTOR shall bear the cost of all other tests, including the retesting of material or workmanship that fails to pass the first test.

The CONTRACTOR shall notify the Engineer in writing, at least 15 days in advance, of its intention to use materials for which tests are specified, to allow sufficient time to perform the tests. The notice shall name the proposed supplier and source of material.

If the notice of intent to use is sent before the materials are available for testing or inspection, or is sent so far in advance that the materials on hand at the time will not last but will be replaced by a new lot prior to use on the Work, it will be the CONTRACTOR's responsibility to renotify the Engineer when samples which are representative may be obtained.

There will be inspection of this project to ensure strict adherence to these specifications. During the course of work, CONTRACTOR shall be responsible for calling the Project Engineer for testing and inspection (48) hours in advance. Work not properly tested and inspected will be subject to rejection.

Any work done in unauthorized areas or in a manner unacceptable to the inspector may not be accepted and/or paid for.

6-1.01B(v) Certification. The Engineer may waive materials testing requirements of the Specifications and accept the manufacturer's written certification that the materials to be supplied meet those requirements. Materials test data may be required as part of the certification.

6-1.01B(vi) Trade Names or Equals. The CONTRACTOR may supply any of the materials specified or offer an equivalent. The Engineer shall determine whether the material offered is equivalent to that specified. Adequate time shall be allowed for the Engineer to make this determination.

Whenever any particular material, process, or equipment is indicated by patent, proprietary or brand name, or by name of manufacturer, such wording is used for the purpose of facilitating its description and shall be deemed to be followed by the words or equal. A listing of materials is not intended to be comprehensive, or in order of preference. The CONTRACTOR may offer any material, process, or equipment considered to be equivalent to that indicated. Approval of equipment and materials offered as equivalents to those specified must be obtained prior to the opening of bids as set forth in the Instructions to Bidders.

The CONTRACTOR shall, at its expense, furnish data concerning items offered by it as equivalent to those specified. The CONTRACTOR shall have the material tested as required by the Engineer to determine that the quality, strength, physical, chemical, or other characteristics, including durability, finish, efficiency, dimensions, service, and suitability are such that the item will fulfill its intended function.

Test methods shall be subject to the approval of the Engineer. Test results shall be reported promptly to the Engineer, who will evaluate the results and determine if the substitute item is equivalent. The Engineer's findings shall be final. Installation and use of a substitute item shall not be made until approved by the Engineer.

If a substitute offered by the CONTRACTOR is not found to be equal to the specified material, the CONTRACTOR shall furnish and install the specified material.

The specified Contract completion time shall not be affected by any circumstance developing from the provisions of this Section.

Along with information supplied by the CONTRACTOR regarding equivalency of the proposed item, the CONTRACTOR shall clearly identify all deviations from the specified item. Deviations discovered by the Engineer after acceptance of an “or equal” item which were not identified by the CONTRACTOR with his/her submittal shall be cause for rejection of the “or equal” item. CONTRACTOR shall be due no additional compensation in time or money for acceptance or rejection of a proposed “or equal” item and subsequent replacement with the item specified. CONTRACTOR shall pay cost to AGENCY for items requiring more than two submittals and analysis of any shop drawing which requires more than a general review of an “or equal” item.

6-1.01B(vii) Weighing and Metering Equipment. All scales and metering equipment used for proportioning materials shall be inspected for accuracy and certified within the past 12 months by the State of California Bureau of Weights and Measures, by the County Director or Sealer of Weights and Measures, or by a scale mechanic registered with or licensed by the County.

The accuracy of the work of a scale service AGENCY, except as stated herein, shall meet the standards of the California Business and Professions Code and the California Code of Regulations pertaining to weighing devices. A certificate of compliance shall be presented, prior to operation, to the Engineer for approval and shall be renewed whenever required by the Engineer at no cost to the AGENCY.

All scales shall be arranged so they may be read easily from the operator’s platform or area. They shall indicate the true net weight without the application of any factor. The figures of the scales shall be clearly legible. Scales shall be accurate to within 1 percent when tested with the plant shut down. Weighing equipment shall be so insulated against vibration or moving of other operating equipment in the plant area that the error in weighing with the entire plant running will not exceed 2 percent for any setting nor 1.5 percent for any batch.

6-1.01B(viii) Calibration of Testing Equipment. Testing equipment, such as, but not limited to pressure gages, metering devices, hydraulic systems, force (load) measuring instruments, and strain-measuring devices shall be calibrated by a testing AGENCY acceptable to the Engineer at intervals not to exceed 12 months and following repairs, modification, or relocation of the equipment. Calibration certificates shall be provided when requested by the Engineer.

7 LEGAL RELATIONS AND RESPONSIBILITY TO THE PUBLIC

7-1.02K(3) Certified Payroll Records (Labor Code § 1776).

Submit certified payroll records to the AGENCY.

7-1.03 PUBLIC CONVENIENCE.

7-1.03A Traffic and Access. The CONTRACTOR’s operations shall cause no unnecessary inconvenience. The access rights of the public shall be considered at all times. Unless otherwise authorized, traffic shall be permitted to pass through the Work, or an approved detour shall be provided.

Safe and adequate pedestrian and vehicular access shall be provided and maintained to: fire hydrants; commercial and industrial establishments; churches, schools and parking lots; service stations and motels; hospitals; police and fire stations; and establishments of similar nature. Access to these facilities shall be continuous and unobstructed unless otherwise approved by the Engineer.

Safe and adequate pedestrian zones and public transportation stops, as well as pedestrian crossings of the Work at intervals not exceeding 90 m (300 feet), shall be maintained unless otherwise approved by the Engineer.

Vehicular access to residential driveways shall be maintained to the property line except when necessary construction precludes such access for reasonable periods of time. If backfill has been completed to the extent that safe access may be provided, and the street is opened to local traffic, the CONTRACTOR shall immediately clear the street and driveways and provide and maintain access.

The CONTRACTOR shall cooperate with the various parties involved in the delivery of mail and the collection and removal of trash and garbage to maintain existing schedules for these services.

Grading operations, roadway excavation and fill construction shall be conducted by the CONTRACTOR in a manner to provide a reasonably satisfactory surface for traffic. When rough grading is completed, the roadbed surface shall be brought to a smooth, even condition satisfactory for traffic.

Unless otherwise authorized, work shall be performed in only one-half of the roadway at one time. One half shall be kept open and unobstructed until the opposite side is ready for use. If one-half a street only is being improved, the other half shall be conditioned and maintained as a detour.

The CONTRACTOR will be required to maintain at least one lane of traffic in each direction through the project area at all times in a manner satisfactory to the Engineer in the form of an engineered traffic control plan. The engineered traffic control plans must be signed by a California registered civil and/or traffic engineer. The plan is a required submittal for review one week prior to the pre-construction meeting.

If traffic control on the project shall be implemented by a sub-contractor, such subcontractor must specialize in Traffic Control and be approved by the City Engineer.

The CONTRACTOR shall include in its Bid all costs for the above requirements.

7-1.03A(i) General. One week prior to pre-construction meeting, the CONTRACTOR shall submit his/her complete construction schedule to the Engineer for approval. The CONTRACTOR shall submit requests for changes in the schedule to the Engineer for approval at least forty-eight (48) hours prior to the scheduled Work.

7-1.03A(ii) Parking Restrictions and Posting for Tow Away. No Parking signs, posted by the CONTRACTOR, shall be of heavy card stock and not less than 1.75 square feet of surface area on the face. Background color shall be white and letters shall be printed in red water resistant ink except day, date, and time of restriction may be printed in black water resistant ink. The signs shall be printed with the words "Tow Away" and "No Parking" with a character height of not less than 2.75 inches and a stroke width of not less than 0.5 inches. The day, dated, and time of the particular restriction shall be printed or attached below the above-mentioned wording in characters of not less than 2.0 inches in height and 0.4 inches in stroke width. The day of the week shall be written

out or properly abbreviated with three to four letters; date or dates or restriction shall be listed completely; the beginning and ending times shall be clearly listed on the sign.

Signs shall be mounted such that the wording “No Parking” are at an elevation at least three feet above the adjacent flowline. Signs may be tied with string to trees and power poles, taped to existing sign poles, or mounted to stakes or barricades as provided by the CONTRACTOR. The signs shall be placed as needed to control the parking of cars within the construction zone; signs shall be placed at intervals of 75 feet or less along each side of the roadway.

Signs shall be posted and maintained by the CONTRACTOR for a period of 72 hours prior to the restrictions becoming effective. The CONTRACTOR may only post parking restrictions that are effective for the duration of the Work. Upon completion of the Work, the CONTRACTOR shall promptly and completely remove and dispose all signs, stakes, and barricades. The CONTRACTOR shall promptly reset or replace all damaged or defective signs.

7-1.03B Storage of Equipment and Materials in Public Streets. Construction materials shall not be stored in streets, roads, or highways for more than 5 days after unloading. All materials or equipment not installed or used in construction within 5 days after unloading shall be stored elsewhere by the CONTRACTOR at its expense unless authorized additional storage time.

Construction equipment shall not be stored at the Work site before its actual use on the Work nor for more than 5 days after it is no longer needed. Time necessary for repair or assembly of equipment may be authorized by the Engineer.

Excavated material, except that which is to be used as backfill in the adjacent trench, shall not be stored in public streets unless otherwise permitted. After placing backfill, all excess material shall be removed immediately from the site.

7-1.03C Street Closures, Detours, Barricades. The CONTRACTOR shall comply with all applicable State, County and City requirements for closure of streets. The CONTRACTOR shall provide barriers, guards, lights, signs, temporary bridges, flagpersons, and watchpersons. The CONTRACTOR shall be responsible for compliance with additional public safety requirements which may arise. The CONTRACTOR shall furnish and install signs and warning devices and promptly remove them upon completion of the Work.

At least 48 hours in advance of closing, partially closing or reopening, any street, alley, or other public thoroughfare, the CONTRACTOR shall notify the Police, Fire, and Public Works Departments, and comply with their requirements. Deviations must first be approved in writing by the Engineer.

The CONTRACTOR shall secure approval, in advance, from authorities concerned for the use of any bridges proposed by it for public use. Temporary bridges shall be clearly posted as to load limit, with signs and posting conforming to current requirements covering “signs” as set forth in the Traffic Manual published by the California Department of Transportation. This manual shall also apply to the street closures, barricades, detours, lights, and other safety devices required.

All traffic control barricades, signs and devices used by the CONTRACTOR shall, as a minimum, conform to the latest edition of the “California Manual on Uniform Traffic Controls Devices” (“MUTCD”). Channelization devices shall be spaced no greater than fifty (50) feet apart. The CONTRACTOR shall take additional precautions as he/she may find necessary under the circumstances.

Should the CONTRACTOR fail to provide adequate traffic control or safety barricades, and in the event a responsible individual cannot be located or refuses to perform, the AGENCY will at its option place needed devices or engage a private firm to place and maintain said barricades, which will be charged to the CONTRACTOR directly.

Full street closures will not be allowed without City Council approval.

All costs involved shall be included in the Bid.

7-1.04 PUBLIC SAFETY.

Payment for performing all work necessary to provide safety measures shall be included in the prices bid for other items of work except where separate bid items for excavation safety are provided, or required by law.

7-1.04A Safety Orders. The CONTRACTOR shall have at the Work site, copies of suitable extracts of: Construction Safety Orders, Tunnel Safety Orders and General Industry Safety Orders issues by the State Division of Industrial Safety. The CONTRACTOR shall comply with provisions of these and all other applicable laws, ordinances, and regulations.

7-1.04B Use of Explosives. Explosive may be used only when authorized in writing by the Engineer, or as otherwise stated in the Specifications. Explosives shall be handled, used, and stored in accordance with all applicable regulations.

The Engineer's approval of the use of explosives shall not relieve the CONTRACTOR from liability for claims caused by blasting operations.

7-1.04C Special Hazardous Substances and Processes. Materials that contain hazardous substances or mixtures may be required on the Work. A Material Safety Data Sheet as described in section 5194 of the California Code of Regulations shall be requested by the CONTRACTOR from the manufacturer of any hazardous products used.

Material usage shall be accomplished with strict adherence to California Division of Industrial Safety requirements and all manufacturer warnings and application instructions listed on the Material Safety Data Sheet and on the product container label.

The CONTRACTOR shall notify the Engineer if a specified product cannot be used under safe conditions.

7-1.04D Confined Spaces.

(a) Confined Space Entry Program. The CONTRACTOR shall be responsible for implementing, administering, and maintaining a confined space entry program (CSEP) in accordance with sections 5156, 5157 and 5158, Title 8, CCR.

Prior to starting the Work, the CONTRACTOR shall prepare and submit its comprehensive CSEP to the Engineer. The CSEP shall address all potential physical and environmental hazards and contain procedures for safe entry into confined spaces, including, but not limited to the following:

1. Training of personnel
2. Purging and cleaning of materials and residue
3. Potential isolation and control of energy and material inflow
4. Controlled access to the space.

5. Atmospheric testing of the space
6. Ventilation of the space
7. Special hazards consideration
8. Personal protective equipment
9. Rescue plan provisions

The CONTRACTOR's submittal shall include the names of its personnel, including subcontractor personnel, assigned to the project who will have CSEP responsibilities, their CSEP training, and their specific assignment and responsibility in carrying out the CSEP.

(b) Permit-Required Confined Spaces. Entry into permit-required confined spaces as defined in section 5157, Title 8, CCR may be required as a part of the Work. All manholes, tanks, vaults, pipelines, excavations, or other enclosed or partially enclosed spaces shall be considered permit-required confined spaces until the pre-entry procedures demonstrate otherwise. The CONTRACTOR shall implement a permit space program prior to performing any work in a permit-required confined space. A copy of the permit shall be available at all times for review by CONTRACTOR and AGENCY personnel at the Work site.

(c) Payment. Payment for implementing, administering, and providing all equipment and personnel to perform the CSEP shall be included in the bid items for which the CSEP is required.

7-1.04E SPECIAL NOTIFICATIONS

Advanced notification shall be given to all businesses and residents adjacent to work.

Section 42 15.25 – 42 17 of the Government code of the State of California (2) working days prior to commencing any excavation "Underground Service Alert of Northern California" (USA) shall be notified by phone, toll-free 1-800-227-2600 for the assignment of an Inquiry Identification Number. **No excavation shall commence unless the CONTRACTOR has obtained the USA Inquiry Identification Number.**

The CONTRACTOR shall contact all utility companies at least five (5) working days prior to commencing work and shall verify the location of any known utilities and determine whether or not a representative of each company will be present during excavation.

The known public utilities are:

City of Auburn (sewer and storm drain):	Paige Culbertson 530-823-4211 ext. 147
Placer County Water Agency (PCWA):	144 Ferguson Road PO Box 6570 Auburn CA 95604 530-823-4850
PG&E (gas and electric):	343 Sacramento Street, Auburn., CA 95603 530-889-3190
AT&T (communications):	2823 Bell Road Auburn, CA 95603 530- 885-7702 800-228-2020
Astound (communications):	1015 Lincoln Way, Auburn CA 95603 866- 928-3123

The existing subsurface utilities have been indicated based on the best available recorded information.

However, to avoid or resolve and interference problems between these existing utilities and the propose work the CONTRACTOR shall field verify the vertical and horizontal locations of all utilities, such water lines and water services, electronic conduits, telephone and television cable, storm drain facilities, and all other facilities and obstructions prior to beginning any excavations. If conflicts exist, revised grades and/or alignments may be established, if required. **Such field verification shall require exposing these utilities prior to the start of construction.**

Additionally, the CONTRACTOR shall also notify the following local entities of his/her schedule fourteen (14) days prior to commencing work, including local refuse collectors, street sweepers, the Post Office, Public Schools, and Bus Companies:

City of Auburn Police Department:	(530) 823-4237
City of Auburn Fire Department:	(530) 8234237
City of Auburn Public Works Department:	(530) 823-4211
CalFire:	(530) 745-3575
Placer County Transit:	(530) 745-7570
Mid-Placer Public School Transportation Agency:	(530) 823-4820
Auburn Union School District:	(530) 885-7242
Placer Union School District:	(530) 886-4400
Recology Auburn Placer:	(530) 885-3735
U.S. Post Office:	(530) 885-7837
Placer County Superior Court:	(530) 745-2000
Gold Country Fairgrounds:	(530) 823-4533

7-1.06 INSURANCE

Specifications for insurance requirements are described in 3-1.07 INSURANCE POLICIES in these Special Provisions.

8 PROSECUTION AND PROGRESS

8-1.01 PRE-CONSTRUCTION MEETING AND SUBMITTAL. A pre-construction meeting will be conducted by the AGENCY prior to commencement of construction at a time and place designated by the AGENCY. Those attending the meeting shall include, but not be limited to, the following:

- a. The CONTRACTOR's representative(s)
- b. Subcontractors' representative(s), if needed
- c. City of Auburn Director of Public Works
- d. City of Auburn Project Manager
- e. The Engineers and Engineering Consultants
- f. The City Engineer
- g. City of Auburn Public Works Inspectors

One week prior to this meeting the CONTRACTOR shall submit the following:

1. Construction Schedule
2. Traffic Control Plan
3. Emergency Contact List
4. List of Subcontractors
5. Storm Water Pollution Prevention Plan (SWPPP) or Erosion and Sediment Control Plan ("ESCP") as applicable

8-1.02 CONSTRUCTION SCHEDULE. After notification of award and prior to start of any work, the CONTRACTOR shall submit its proposed construction schedule to the Engineer for approval. The construction schedule shall be in the form of a tabulation, chart, or graph and shall be in sufficient detail to show chronological relationship of all activities of the project. These include but are not limited to: estimated starting and completion dates of various activities, submittal of shop drawings to the Engineer for approval, procurement of materials and scheduling of equipment. The construction schedule shall recognize the requirements of 8-1.08 and reflect completion of all work under the Contract within the specified time and in accordance with the Specifications.

Unless otherwise provided, the Contract time shall commence upon the date of issuance of a notice to proceed. The Work shall start within 15 days thereafter and be diligently prosecuted to completion within the time provided in the Specifications.

If the CONTRACTOR desires to make a major change in the method of operations after commencing construction, or if the schedule fails to reflect the actual progress, the CONTRACTOR shall submit to the Engineer a revised construction schedule in advance of beginning revised operations.

The Engineer may waive these requirements for work constructed under permit.

Prior to issuing the Notice to Proceed, the Engineer will schedule and conduct a pre-construction meeting with the CONTRACTOR to review the proposed construction schedule and delivery dates, arrange utility coordination, discuss construction methods, and clarify inspection procedures.

8-1.03 EMERGENCY CONTACT LIST. The CONTRACTOR shall provide the following information in writing and submit it with the signed contract, contract bonds and certificates of insurance. Failure to comply may result in delays in the processing of the contract documents.

1. Name of authorized representative at the job site.
2. Address and telephone number where the above person can be reached 24 hours a day.
3. Address of the nearest office of the CONTRACTOR, if any, and the name and telephone number of a person at that office who is familiar with the project.
4. Address and telephone number of the CONTRACTOR's main office and the name and telephone number of the person at that office familiar with the project.

8-1.04 PROSECUTION OF WORK. To minimize public inconvenience and possible hazard and to restore street and other work areas to their original condition and state of usefulness as soon as practicable, the CONTRACTOR shall diligently prosecute the Work to completion. If the Engineer

determines that the CONTRACTOR is failing to prosecute the Work to the proper extent, the CONTRACTOR shall, upon orders from the Engineer, immediately take steps to remedy the situation. All costs of prosecuting the Work as described herein shall be included in the CONTRACTOR's Bid. Should the CONTRACTOR fail to take the necessary steps to fully accomplish said purposes, after orders of the Engineer to do so, the AGENCY may suspend the work in whole or in part, until the CONTRACTOR takes said steps at no cost to the AGENCY.

As soon as possible under the provisions of the Specifications, the CONTRACTOR shall backfill all excavations and restore to usefulness all improvements existing prior to the start of the Work.

If Work is suspended through no fault of the AGENCY, all expenses and losses incurred by the CONTRACTOR during such suspensions shall be borne by the CONTRACTOR. If the CONTRACTOR fails to properly provide for public safety, traffic, and protection of the Work during periods of suspension, the AGENCY may elect to do so, and deduct the cost thereof from monies due the CONTRACTOR. Such actions will not relieve the CONTRACTOR from liability.

The CONTRACTOR shall submit monthly progress reports to the Engineer by the tenth day of each month. The report shall include an updated construction schedule. Any deviations from the original schedule shall be explained. Progress payments will be withheld pending receipt of any outstanding reports.

8-1.04A Advertising. The names, addresses, phone number and specialties of CONTRACTOR, Subcontractors, architects, or engineers shall be displayed on removable signs. The size and location shall be subject to the Engineer's approval.

Commercial advertising matter shall not be attached to or painted on the surfaces of buildings, fences, canopies, or barricades.

8-1.04B Laws to be Observed. The CONTRACTOR shall keep fully informed of State and National laws and County and Municipal ordinances and regulations which in any manner affect those employed in the Work or the materials used in the Work or in any way affect the conduct of the Work. The CONTRACTOR shall at all times observe and comply with such laws, ordinances, and regulations.

8-1.05 SUSPENSION OF WORK.

8-1.05A General. The Work may be suspended in whole or in part when determined by the Engineer that the suspension is necessary in the interest of the AGENCY. The CONTRACTOR shall comply immediately with any written order of the Engineer. Such suspension shall be without liability to the CONTRACTOR on the part of the AGENCY except as otherwise specified in 8-1.08C.

If the performance of all or any portion of the work is suspended or delayed by the Engineer in writing for an unreasonable period of time (not originally anticipated, customary, or inherent to the construction industry) and the CONTRACTOR believes that additional compensation and/or Contract time is due as a result of such suspension or delay, the CONTRACTOR shall submit to the Engineer in writing a request for adjustment within seven (7) calendar days of receipt of the notice to resume work. The request shall set forth the reasons and support for such adjustment.

Upon receipt, the Engineer will evaluate the CONTRACTOR's request. If the Engineer agrees that the cost and/or time required for the performance of the Contract has increased as a result of such suspension and the suspension was caused by conditions beyond the control of and not the fault of

the CONTRACTOR, its suppliers, or subcontractors at any approved tier, and not caused by weather, the Engineer will make an adjustment (excluding profit) and modify the Contract in writing accordingly. The CONTRACTOR will be notified of the Engineer's determination whether or not an adjustment of the Contract is warranted.

No Contract adjustment will be allowed unless the CONTRACTOR has submitted the request for adjustment within the time prescribed.

No Contract adjustment will be allowed under this clause to the extent that performance would have been suspended or delayed by any other cause, or for which an adjustment is provided or excluded under any other term or condition of this Contract.

8-1.05B Archaeological and Paleontological Discoveries. If discovery is made of items of archaeological or paleontological interest, the CONTRACTOR shall immediately cease excavation in the area of discovery and shall not continue until ordered by the Engineer. When resumed, excavation operations within the area of discovery shall be as directed by the Engineer.

Discoveries which may be encountered may include, but not be limited to, dwelling sites, stone implements or other artifacts, animal bones, human bones, and fossils.

The CONTRACTOR shall be entitled to an extension of time and compensation in accordance with the provisions of 8-1.08.

8-1.06 DEFAULT BY CONTRACTOR.

If the CONTRACTOR fails to begin delivery of material and equipment, to commence the Work within the time specified, to maintain the rate of delivery of material, to execute the Work in the manner and at such locations as specified, or fails to maintain the Work schedule which will insure the AGENCY's interest, or, if the CONTRACTOR is not carrying out the intent of the Contract, the AGENCY may serve written notice upon the CONTRACTOR and the Surety on its Faithful Performance Bond demanding satisfactory compliance with the Contract.

The Contract may be canceled by the City Council without liability for damage, when in the City Council's opinion, the CONTRACTOR is not complying in good faith, has become insolvent, has breached any provision of the Contract, or has assigned or subcontracted any part of the Work without the City Council's consent. In the event of such cancellation, the CONTRACTOR will be paid the actual amount due based on Contract Unit Prices or lump sums bid and the quantity of the Work completed at the time of cancellation, less damages caused to the AGENCY by acts of the CONTRACTOR. The CONTRACTOR, in having tendered a Bid, shall be deemed to have waived any and all claims for damages because of cancellation of Contract for any such reason. If the AGENCY declares the Contract canceled for any of the above reasons, written notice to that effect shall be served upon the Surety. The Surety shall, within 5 days, assume control and perform the Work as successor to the CONTRACTOR.

If the Surety assumes any part of the Work, it shall take the CONTRACTOR's place in all respects for that part, and shall be paid by the AGENCY for all work performed by it in accordance with the Contract. If the Surety assumes the entire Contract, all money due the CONTRACTOR at the time of its default shall be payable to the Surety as the Work progresses, subject to the terms of the Contract.

If the Surety does not assume control and perform the Work within 5 days after receiving a notice of cancellation, or fails to continue to comply, the AGENCY may exclude the Surety from the premises. The AGENCY may then take possession of all material and equipment and complete the Work by AGENCY forces, by letting the unfinished Work to another CONTRACTOR, or by a combination of such methods. In any event, the cost of completing the Work shall be charged against the CONTRACTOR and its Surety and may be deducted from any money due or becoming due from the AGENCY. If the sums due under the Contract are insufficient for completion, the CONTRACTOR or Surety shall pay to the AGENCY within 5 days after the completion, all costs in excess of the sums due.

The provisions of this Subsection shall be in addition to all other rights and remedies available to the AGENCY under law.

8-1.07 TERMINATION OF THE CONTRACT.

The City Council may terminate the Contract at its own discretion or when conditions encountered during the Work make it impossible or impracticable to proceed, or when the AGENCY is prevented from proceeding with the Contract by act of God, by law, or by official action of a public authority.

The City Council may also terminate performance of work under this Contract in whole or, from time to time, in part if the City Council determines that a termination is in the AGENCY's best interest. The AGENCY may terminate any resulting Contract for convenience by providing, (1) a statement that the Contract is being terminated for the convenience of the AGENCY, (2) the effective date of termination, (3) the extent of termination, (4) any special instructions, and (5) the steps the Contractor is to take to minimize the impact on AGENCY.

Upon any notification of termination for convenience, the Contractor is to, (1) stop work immediately on the terminated portion of the Contract, (2) terminate all subcontracts related to the terminated portion of the Contract, (3) advise the AGENCY of any special circumstances precluding stoppage of work, (4) perform the continued portion of the Contract if the termination is partial, (5) take any action necessary to protect property in the Contractor's possession in which the AGENCY has an interest, (6) notify the AGENCY of any legal proceedings growing out of any subcontract, (7) settle any subcontractor claims arising out of the termination, and (8) dispose of termination inventory as directed by the AGENCY.

8-1.08 DELAYS AND EXTENSIONS OF TIME.

8-1.08A General. If delays are caused by unforeseen events beyond the control of the CONTRACTOR, such delays will entitle the CONTRACTOR to an extension of time as provided herein, but the CONTRACTOR will not be entitled to damages or additional payment due to such delays, except as provided in 8-1.08C. Such unforeseen events may include: war, government regulations, labor disputes, strikes, fires, floods, adverse weather or elements necessitating cessation of work, inability to obtain materials, labor or equipment, required extra work, or other specific events as may be further described in the Specifications.

No extension of time will be granted for a delay caused by the CONTRACTOR's inability to obtain materials unless the CONTRACTOR furnishes to the Engineer documentary proof. The proof must be provided in a timely manner in accordance with the sequence of the CONTRACTOR's operations and the approved construction schedule.

If delays beyond the CONTRACTOR's control are caused by events other than those mentioned above, the Engineer may deem an extension of time to be in the best interests of the AGENCY. The CONTRACTOR will not be entitled to damages or additional payment due to such delays, except as provided in 8-1.08C.

If delays beyond the CONTRACTOR's control are caused solely by action or inaction by the AGENCY, such delays will entitle the CONTRACTOR to an extension of time as provided in 8-1.08B(i).

8-1.08A(i) Notice of Delays. Whenever the CONTRACTOR foresees any delay in the prosecution of the work, and in any event immediately upon the occurrence of any delay which the CONTRACTOR regards as unavoidable, he/she shall notify the Engineer in writing of the probability of the occurrence of such delay and its cause so that the Engineer may take immediate steps to prevent, if possible, the occurrence or continuance of the delay, or, if prevention is not possible, may determine whether the delay is to be considered avoidable or unavoidable, how long it continues, and to what extent it will delay the prosecution and completion of the work. It will be concluded that any and all delays which have occurred in the prosecution and completion of the work have been avoidable delays, except such delays as shall have been called to the attention of the Engineer at the time of their occurrence and found by him/her to have been unavoidable. The CONTRACTOR shall make no claims for any delay not called to the attention of the Engineer at the time of its occurrence as an unavoidable delay.

8-1.08A(ii) Avoidable Delays. Avoidable delays in the prosecution or completion of the work shall include all delays which in the opinion of the Engineer would have been avoided by the exercise of care, prudence, foresight, and diligence on the part of the CONTRACTOR of his/her subcontractors. The following shall be considered avoidable delays within the meaning of the contract: 1) Delays in the prosecution of parts of the work which may in themselves be unavoidable but do not necessarily prevent or delay the prosecution of other parts of the work or the completion of the whole work within the time herein specified; 2) Reasonable loss of time resulting from the necessity of submitting samples of materials and drawings to the Engineer for approval and from performing tests of materials, measurements, and inspections; 3) Reasonable interference of other contractors employed by the AGENCY and/or other contractors working in the area which do not necessarily prevent the completion of the whole work within the time agreed upon; 4) Delays resulting from inaccurate or incomplete shop drawing submittals; and 5) Interference of other contractors performing concurrent work.

8-1.08A(iii) Extension of Time. In case the work is not completed in the time specified, including such extensions of time as may have been granted for unavoidable delays, the CONTRACTOR will be assessed damages for delay in accordance with 8-1.11. The AGENCY, however, shall have the right to grant an extension of time for avoidable delay if it is deemed in his/her best interest to do so. During such extension of time, the CONTRACTOR will be charged for engineering and inspection services and other costs as provided 8-1.08B(i) but will not be assessed damages pursuant to 8-1.11.

The CONTRACTOR may make a Claim for an extension of the Contract time, for an Excusable Delay or a Compensable Delay, as defined in Section 8-1.08C below, subject to the following:

(a) If an Excusable Delay and a Compensable Delay occur concurrently, the maximum extension of the Contract time shall be the number of days from the commencement of the first delay to the cessation of the delay which ends last.

(b) If an Unexcusable Delay occurs concurrently with either an Excusable Delay or a Compensable Delay, the maximum extension of the Contract time shall be the number of days, if any, by which the Excusable Delay or the Compensable Delay exceeds the Unexcusable Delay.

(c) If an Unexcusable Delay occurs concurrently with both an Excusable Delay and a Compensable Delay, the maximum extension in the Contract time shall be the number of days, if any, by which the number of days determined pursuant to Subparagraph (a) exceeds the number of days of the Unexcusable Delay.

8-1.08B Extensions of Time. Extensions of time, when granted, will be based upon the effect of delays to the Work, they will not be granted for noncontrolling delays to minor portions of the Work unless it can be shown that such delays did or will delay the progress of the Work.

8-1.08C Payment for Delays to CONTRACTOR. As used herein, the following terms shall have the following meanings:

"Excusable Delay" means any delay of the completion of the Project beyond the expiration of the Contract time caused by conditions beyond the control and without the fault or negligence of the CONTRACTOR such as strikes, embargoes, fire, unavoidable casualties, unusual delays in transportation, national emergency, and stormy and inclement weather conditions in which the work cannot continue. The financial inability of the CONTRACTOR or any subcontractor and default of any subcontractor, without limitation, shall not be deemed conditions beyond the CONTRACTOR's control. An Excusable Delay may entitle the CONTRACTOR to an adjustment in the Contract time.

"Compensable Delay" means any delay of the completion of the work beyond the expiration date of the Contract time caused by the gross negligence or willful acts of the AGENCY, and which delay is unreasonable under the circumstances involved, and not within the contemplation of the parties. A Compensable Delay may entitle the CONTRACTOR to an extension of the Contract time or Contract price. Except as provided herein, the CONTRACTOR shall have no claim for damage or compensation for any delay, interruption, hinderance, or disruption.

"Unexcusable Delay" means any delay of the completion of the Project beyond the expiration of the Contract time resulting from causes other than those listed above. An Unexcusable Delay shall not entitle the CONTRACTOR to an extension of the Contract time or an adjustment of the Contract price.

For a Compensable Delay, the CONTRACTOR shall only be entitled to an adjustment in the Contract price in an amount equal to the actual additional labor costs, material costs, and unavoidable equipment costs incurred by the CONTRACTOR as a result of the Compensable Delay, plus the actual additional wages or salaries and fringe benefits and payroll taxes of supervisory and administrative personnel necessary and directly employed at the Project site for the supervision of the work during the period of Compensable Delay. Except as provided herein, the CONTRACTOR shall have no claim for damage or compensation for any delay, interruption, hinderance, or disruption. There shall be no Compensable Delay unless the event or occurrence giving rise to the Compensable Delay extends the actual completion of the Project past the Contract time.

The parties agree that the AGENCY's exercise of its rights to order changes in the work, regardless of the extent and number of changes, or to suspend the work, is within the contemplation of the parties and shall not be the basis for any Claim for Compensable Delay. The rights of the

CONTRACTOR to adjustments of the Contract Time and the Contract Sum, based on changes ordered in the work or suspension of the work, shall be solely governed by this provision.

8-1.09 TIME OF COMPLETION

8-1.09A General. The CONTRACTOR shall complete the Work within the time set forth in the Contract. The CONTRACTOR shall complete each portion of the Work within such time as set forth in the Contract for such portion. Unless otherwise specified, the time of completion of the Contract shall be expressed in working days.

8-1.09B Working Day. A working day is any day within the period between the start of the Contract time and the date provided for completion, as defined in Section B – INSTRUCTION TO BIDDERS (B1.03), or upon field acceptance by the Engineer for all work provided for in the Contract, whichever occurs first, other than:

1. Saturday,
2. Sunday,
3. any day designated as a holiday by the AGENCY,
4. any other day designated as a holiday in a Master Labor Agreement entered into by the CONTRACTOR or on behalf of the CONTRACTOR as an eligible member of a contractor association,
5. any day the CONTRACTOR is prevented from working at the beginning of the workday for cause as defined in 8-1.08,
6. any day the CONTRACTOR is prevented from working during the first 5 hours with at least 60 percent of the normal work force for cause as defined in 8-1.08.

The CONTRACTOR's activities involving work which requires street closure, detours, and barricades shall be confined to the hours between 9:00 a.m. and 6:00 p.m. Monday through Friday. In addition, the CONTRACTOR shall not perform any Work on Saturday, Sunday, or on AGENCY-designated holidays. AGENCY-designated holidays are listed in TABLE 1 – AGENCY-DESIGNATED HOLIDAYS below. Deviation from these hours will be permitted upon approval of the Engineer, except in emergencies involving immediate hazard to persons or property.

Deviations from these hours will not be permitted without the prior consent of the Engineer, except in emergencies involving immediate hazard to persons or property. In the event of either a requested or emergency deviation, inspection service fees will be charged against the CONTRACTOR. Service fees will be calculated at overtime rates including benefits, overhead, and travel time; and will be deducted from the amounts due the CONTRACTOR.

Failure of the CONTRACTOR to adhere to working day requirements will result in damages being sustained by the AGENCY. Such damages are, and will continue to be, impracticable and extremely difficult to determine.

TABLE 1 – AGENCY-DESIGNATED HOLIDAYS

January 1	New Year's Day
Third Monday in January	Martin Luther King Day
Third Monday in February	Presidents' Day
Last Monday in May	Memorial Day
June 19	Juneteenth
July 4	Independence Day
First Monday in September	Labor Day
November 11	Veterans Day
Fourth Thursday in November	Thanksgiving Day
Fourth Friday in November	Day after Thanksgiving Day
December 24	Christmas Eve
December 25	Christmas Day

8-1.09C Contract Time Accounting. The Engineer will make a daily determination of each working day to be charged against the Contract time. These determinations will be discussed, and the CONTRACTOR will be furnished a periodic statement showing allowable number of working days of Contract time, as adjusted, at the beginning of the reporting period. The statement will also indicate the number of working days charged during the reporting period and the number of working days of Contract time remaining. If the CONTRACTOR does not agree with the statement, it shall file a written protest within 15 days after receipt, setting forth the facts of the protest. Otherwise, the statement will be deemed to have been accepted.

8-1.10 COMPLETION, ACCEPTANCE, AND WARRANTY. The Work will be inspected by the Engineer for acceptance upon receipt of the CONTRACTOR's written assertion that the Work has been completed.

If, in the Engineer's judgment, the Work has been completed and is ready for acceptance, it will so certify to the City Council, which may accept the completed Work. The Engineer will, in its certification to the City Council, give the date when the Work was completed. This will be the date when the CONTRACTOR is relieved from responsibility to protect the Work.

All work shall be warranted by the CONTRACTOR against defective workmanship and materials for a period of 1 year from the date a Notice of Completion is filed. The CONTRACTOR shall replace or repair any such defective work in a manner satisfactory to the Engineer, after notice to do so from the Engineer, and within the time specified in the notice. If the CONTRACTOR fails to make such replacement or repairs within the time specified in the notice, the AGENCY may perform this work and the CONTRACTOR's sureties shall be liable for the cost thereof.

8-1.10A General Guaranty. The CONTRACTOR shall remedy any defects in the Work and pay for any damage to other work resulting therefrom, which shall appear within a period of one year from the date of final acceptance of the Work unless a longer period is specified. The AGENCY will give notice of observed defects with reasonable promptness.

8-1.11 LIQUIDATED DAMAGES. The CONTRACTOR shall complete all or any designated portion of the Work called for under the Contract within the time set forth in Section B – INSTRUCTION TO BIDDERS (B1.03).

In accordance with Government Code 53069.85, and all other applicable law, the CONTRACTOR agrees to forfeit and pay the AGENCY the amount of Five Hundred Dollars (\$500.00) per day for each and every day of unauthorized delay beyond the completion date, which shall be deducted from any monies due the CONTRACTOR. This payment shall be considered liquidated damages. CONTRACTOR agrees that such liquidated damages are reasonable under the circumstances existing at the time of execution of the contract, that such liquidated damages are to compensate AGENCY for losses that are difficult to measure and that such damages are not a penalty.

Failure of the CONTRACTOR to perform any covenant or condition contained in the Contract Documents within the time period specified shall constitute a material breach of this Contract entitling the AGENCY to terminate the Contract unless the CONTRACTOR applies for, and receives, an extension of time in accordance with the procedures set forth in 8-1.08.

Failure of the AGENCY to insist upon the performance of any covenant or conditions within the time period specified in the Contract Documents shall not constitute a waiver of the CONTRACTOR's duty to complete performance within the designated periods unless the AGENCY has executed a waiver in writing.

The AGENCY's agreement to waive a specific time provision or to extend the time for performance shall not constitute a waiver of any other time provision contained in the Contract Documents.

Failure of the CONTRACTOR to complete performance promptly within the additional time authorized in a waiver or extension of time agreement shall constitute a material breach of this Contract entitling the AGENCY to terminate this agreement.

The CONTRACTOR shall not be deemed in breach of this Contract and no forfeiture due to delay shall be made because of any delays in the completion of the Work due to unforeseeable causes beyond the control and without the fault or negligence of the CONTRACTOR provided the CONTRACTOR requests an extension of time in accordance with the procedures set forth in 8-1.08. Unforeseeable causes of delay beyond the control of the CONTRACTOR shall include acts of God, acts of a public enemy, acts of the government, acts of the AGENCY, or acts of another CONTRACTOR in the performance of a contract with the AGENCY, fires, floods, epidemics, quarantine restrictions, strikes, freight embargoes, and weather, or delays of subcontractors due to such causes, or delays caused by failure of the owner of a utility to provide for removal or relocation of existing utility facilities. Delays caused by actions or neglect of CONTRACTOR or his/her agents, servants, employees, officers, subcontractors, directors, or of any party contracting to perform part of all the Work or to supply any equipment or materials shall not be excusable delays. Excusable delays (those beyond the CONTRACTOR's control) shall not entitle the CONTRACTOR to any additional compensation. The sole recourse of the CONTRACTOR shall be to seek an extension of time.

8-1.12 DISPUTES AND CLAIMS

8-1.12A General.

A Claim is any request by CONTRACTOR to adjust, alter, modify, or otherwise change the Contract price or the Contract time, or both. A Claim must be stated with specificity, including

identification of the event or occurrence giving rise to the Claim, the date of the event, and the asserted effect on the Contract price and the Contract time, if any. The Claim shall include adequate supporting data. Adequate supporting data for a Claim for an adjustment of the Contract time shall include scheduling data demonstrating the impact of the event on the controlling operation and completion of the Project. Adequate supporting data for a Claim for an adjustment in the Contract price shall include a detailed cost breakdown of items included within the Claim and documentation supporting each item of cost.

Any and all decisions made on appeal pursuant to this Subsection shall be in writing. Any “decision” purportedly made pursuant to this Subsection that is not in writing shall not be binding upon the AGENCY and should not be relied upon by the CONTRACTOR.

Nothing in this Subsection shall be considered as relieving the CONTRACTOR from his/her duty to file the notice required under 8-1.13 or other duties required by the Contract Documents.

8-1.12B Records Supporting Claims

The CONTRACTOR and all subcontractors shall keep full and complete records of the costs and additional time incurred for any work for which a claim for additional compensation is made. The Engineer or any designated claim investigator or auditor shall have access to those records and any other records as may be required by the Engineer to determine the facts or contentions involved in the claims. CONTRACTOR agrees that failure to permit access to those records waives CONTRACTOR’s claims.

The AGENCY, or its authorized representatives, shall have access, upon reasonable notice, during normal business hours, to CONTRACTOR and subcontractors’ books, documents and accounting records, including but not limited to, bid worksheets, bids, subcontractor bids and proposals, estimates, cost accounting data, accounting records, payroll records, time sheets, canceled checks, profit and loss statements, balance sheets, project correspondence including but not limited to all correspondence between CONTRACTOR and its sureties and subcontractors/vendors, project files, scheduling information, and other records of the CONTRACTOR and all subcontractors directly or indirectly pertinent to the work, original as well as change and claimed extra work, to verify and evaluate the accuracy of cost and pricing data submitted with any change order, prospective or completed, or any claim for which additional compensation has been requested or claim has been tendered. Such access shall include the right to examine and audit such records, and make excerpts, transcriptions, and photocopies at AGENCY’s cost.

The CONTRACTOR and all subcontractors shall maintain records in such a manner as to provide a clear distinction between the direct costs of extra work paid and the costs of other operations.

From the above records, the CONTRACTOR shall furnish the Engineer completed daily extra work reports, either on forms furnished by the AGENCY or on computerized facsimiles of the AGENCY's forms acceptable to the Engineer, for each day's extra work to be paid for on a force account basis. The daily extra work reports shall itemize the materials used, and shall state the direct cost of labor and the charges for equipment rental, whether furnished by the CONTRACTOR, subcontractor or other forces. The daily extra work reports shall provide names or identifications and classifications of workers, the hourly rate of pay and hours worked, and also the size, type and identification number of equipment, and hours operated.

Material charges shall be substantiated by valid copies of vendor's invoices. The invoices shall be submitted with the daily extra work reports, or if not available, the invoices shall be submitted

within 30 days after the date of delivery of the material or within 15 days after the acceptance of the Contract, whichever occurs first. CONTRACTOR waives payment for material charges not substantiated by valid copies of vendor's invoices submitted within the times provided.

Daily extra work reports shall be signed by the CONTRACTOR or the CONTRACTOR's authorized representative.

The Engineer will compare the Engineer's records with the completed daily extra work reports furnished by the CONTRACTOR and make any necessary adjustments. When these daily extra work reports are agreed upon and signed by both parties, the reports shall become the basis of payment for the work performed, but shall not preclude subsequent adjustment based on a later audit by the City.

The CONTRACTOR's and all subcontractor's records pertaining to the Project shall be open to inspection or audit by representatives of the AGENCY, during the life of the Contract and for a period of not less than 3 years after the date of acceptance thereof, and the CONTRACTOR and all subcontractors shall retain those records for that period. Where payment for materials or labor is based on the cost thereof to forces other than the CONTRACTOR, the CONTRACTOR shall make every reasonable effort to ensure that the cost records of those other forces will be open to inspection and audit by representatives of the AGENCY on the same terms and conditions as the cost records of the CONTRACTOR.

If an audit is to be commenced more than 60 days after the acceptance date of the Contract, the CONTRACTOR will be given a reasonable notice of the time when the audit is to begin.

8-1.12C Administrative Review. Request for review made to the Construction Inspector or Project Engineer may be either oral or written. Request for review made to the City Engineer shall be made in writing with supporting evidence attached.

The CONTRACTOR shall submit each request for review within twenty-one (21) calendar days of receipt of the decision that he/she is requesting.

Prior to demand for arbitration, the CONTRACTOR shall exhaust his/her administrative remedies by attempting to resolve his/her dispute or claim with City Engineer

The City Engineer shall address disputes or claims within twenty-eight (28) calendar days after receiving such request and all necessary supporting data. The City Engineer's decision on the dispute or claim shall be the AGENCY's final decision.

8-1.13 NOTICE OF POTENTIAL CLAIM. The CONTRACTOR shall not be entitled to the payment of any additional compensation for any cause, including any act, or failure to act, by the Engineer, or the happening of any event, thing or occurrence, unless the CONTRACTOR shall have given the Engineer due notice in writing, of the potential claim as hereinafter specified, provided, however, that compliance with this Subsection shall not be a prerequisite as to any claim that is based on differences in measurements or errors of computation as to the Contract quantities.

Additionally, this Subsection shall not supersede the specific notice and protest requirements of Subsection 4-1.05D "Changed Conditions" and Subsection 8-1.09C "Contract Time Accounting" respectively.

A written notice of potential claim shall set forth the reasons the CONTRACTOR believes additional compensation will or may be due, the nature of the costs involved, and, insofar as possible, the amount of the potential claim. A notice as above required must have been given to

the Engineer prior to the time that the CONTRACTOR shall have performed the Work giving rise to the potential claim for additional compensation, if based on an act or failure to act by the Engineer, or in all other cases within fifteen (15) days after the happening of the event, thing or occurrence giving rise to the potential claim. CONTRACTOR hereby expressly waives any Claims of which CONTRACTOR was aware, whether or not the exact amounts of such Claims were ascertainable, and that are not submitted to the AGENCY prior to CONTRACTOR proceeding to perform the work, or portions of the work, giving rise to such Claims.

CONTRACTOR hereby expressly waives all Claims not submitted, in complete and proper form, on or before the date of final payment.

CONTRACTOR expressly waives any Claims for delay or adjustment to the Contract time if, within three (3) days of the event or occurrences giving rise to the delay, the CONTRACTOR fails to provide written notice to AGENCY. Said written notice shall include the event or occurrence giving rise to the delay, the estimated duration of the delay, and the impact of the event or occurrence upon the critical path, controlling operation and completion of the Project.

It is the intention of this Subsection that differences between the parties arising under and by the virtue of the Contract be brought to the attention of the Engineer at the earliest possible time in order that such matters may be settled, if possible, or other appropriate action promptly taken. The CONTRACTOR hereby agrees that he/she shall have no right to additional compensation for any claim that may be based on any such act, failure to act, event, thing or occurrence for which no written notice of potential claim as herein required was filed.

8-1.13 CERTIFICATION OF CLAIMS

California Penal Code section 72, provides that any person who presents for payment with intent to defraud any agency board or officer, any false or fraudulent claim, bill, account, voucher, or writing, is punishable by fines not exceeding ten thousand dollars (\$10,000.00) and/or imprisonment in the state prison.

Government Code sections 12650, et seq., pertains to civil penalties that may be recovered from persons (including corporations, etc.) for presenting a false claim for payment or approval, presenting a false record or statement to get a false claim paid or approved, or other acts, to any officer or employee of any political subdivision of the State of California. Any person or corporation violating the provisions of Government Code sections 12650, et seq., shall be liable for three times the amount of the damages of the political subdivision, plus a civil penalty, plus costs.

All Claims by CONTRACTOR, shall include the following certification, properly completed and executed by CONTRACTOR or an officer of CONTRACTOR:

I, _____, BEING THE _____ (MUST BE AN OFFICER) OF _____ (CONTRACTOR), DECLARE UNDER PENALTY OF PERJURY UNDER THE LAWS OF THE STATE OF CALIFORNIA, AND DO PERSONALLY CERTIFY AND ATTEST THAT: I HAVE THOROUGHLY REVIEWED THE ATTACHED CLAIM FOR ADDITIONAL COMPENSATION AND/OR EXTENSION OF TIME, AND KNOW ITS CONTENTS, AND SAID CLAIM IS TRUTHFUL AND ACCURATE; THAT THE AMOUNT REQUESTED ACCURATELY REFLECTS THE CONTRACT ADJUSTMENT FOR WHICH THE OWNER IS LIABLE; AND, FURTHER, THAT I AM FAMILIAR WITH CALIFORNIA PENAL CODE SECTION 72 AND CALIFORNIA GOVERNMENT CODE SECTION 12650,

List of Approved Key Personnel for N66604-26-R-0163

The following Key Personnel are approved for this task order. Any changes to the below list require approval in accordance with clause C-237-H002 "SUBSTITUTION OF KEY PERSONNEL".

[illegible]

ITEM 2 — PROPERTIES

As shown in the following table, we have more than 450 manufacturing, distribution and service branch or office facilities worldwide. We own the majority of our manufacturing and larger distribution facilities. We lease certain manufacturing facilities and most of our smaller distribution outlets and financial service branches and offices.

Dana Facilities by Geographic Region

Type of Facility	North America	Europe	South America	Asia/ Pacific	Total
Manufacturing	166	70	39	11	286
Distribution	42	17	39	3	101
Service branches, offices	51	12	5	4	72
Total	259	99	83	18	459

ITEM 3 — LEGAL PROCEEDINGS

We are a party to various pending judicial and administrative proceedings arising in the ordinary course of business. After reviewing the proceedings that are currently pending (including the probable outcomes, reasonably anticipated costs and expenses, availability and limits of our insurance coverage, and our established reserves for uninsured liabilities), we do not believe that any liabilities that may result from these proceedings are reasonably likely to have a material adverse effect on our liquidity, financial condition or results of operations.

Under the rules of the Securities and Exchange Commission, we are required to report certain environmental proceedings involving governmental agencies that are not deemed to be routine proceedings incidental to our business. We are not currently a party to any such proceedings.

ITEM 4 — SUBMISSION OF MATTERS TO A VOTE OF SECURITY HOLDERS

- None -

PART II

ITEM 5 — MARKET FOR REGISTRANT’S COMMON EQUITY AND RELATED STOCKHOLDER MATTERS

Our common stock is listed on the New York Stock Exchange and Pacific Exchange. On February 16, 2001, there were 35,620 shareholders of record.

Dividends have been paid on our common stock every year since 1936. Quarterly dividends have been paid since 1942.

You can find more information in “Shareholders’ Investment” on page 50 of our 2000 Annual Report.

ITEM 6 — SELECTED FINANCIAL DATA

You can find selected financial data related to Dana in “Financial Highlights” under “Eleven-Year History” on page 51 of our 2000 Annual Report.

ITEM 7 — MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

You can find “Management’s Discussion and Analysis of Financial Condition and Results of Operations” on pages 40-46 of our 2000 Annual Report.

ITEM 7A – QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

You can find market risk information in “Financial Instruments,” “Derivative Financial Instruments” and “Marketable Securities” under “Note 1. Summary of Significant Accounting Policies” on page 27, in “Note 7. Interest Rate Agreements” on page 29 and in “Note 16. Fair Value of Financial Instruments” on page 37 of our 2000 Annual Report.

ITEM 8 — FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

You can find our financial statements and the report by PricewaterhouseCoopers LLP dated February 2, 2001, on pages 21-39 and “Unaudited Quarterly Financial Information” under “Shareholders’ Investment” on page 50 of our 2000 Annual Report.

ITEM 9 — CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

- None -

PART III

ITEM 10 — DIRECTORS AND EXECUTIVE OFFICERS OF THE REGISTRANT

You can find general information about our directors under “Election of Directors” on pages 1-3 in our 2001 Proxy Statement and information about our executive officers in Part I, Item 1 of this Form 10-K.

You can find information about the filing of reports by our directors, officers and 10% stockholders under Section 16(a) of the Securities Exchange Act of 1934 under “Section 16(a) Beneficial Ownership Reporting Compliance” on page 17 in our 2001 Proxy Statement.

ITEM 11 — EXECUTIVE COMPENSATION

You can find information about executive compensation in the following sections of our 2001 Proxy Statement: “Compensation” on page 4 under “The Board and its Committees,” “Executive Compensation” on pages 7-12 and “Compensation Committee Report on Executive Compensation” on pages 13-15.

You can find information about our stock performance under “Comparison of Five-Year Cumulative Total Return” on page 16 of our 2001 Proxy Statement.

ITEM 12 — SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT

You can find information about the stock ownership of our directors, officers and 5% stockholders under “Stock Ownership” on pages 5-6 of our 2001 Proxy Statement.

ITEM 13 — CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS

You can find information about transactions between Dana and our directors, officers and 5% stockholders under “Other Transactions” on pages 16-17 of our 2001 Proxy Statement.

PART IV

ITEM 14 — EXHIBITS, FINANCIAL STATEMENT SCHEDULES, AND REPORTS ON FORM 8-K

		Annual Report Pages
(a)	The following documents are filed as part of this report:	
(1)	Financial Statements:	
	Report of Independent Accountants	21
	Statement of Income for each of the three years in the period ended December 31, 2000	22
	Balance Sheet at December 31, 1999 and 2000	23
	Statement of Cash Flows for each of the three years in the period ended December 31, 2000	24
	Statement of Shareholders' Equity for each of the three years in the period ended December 31, 2000	25
	Notes to Financial Statements	26 - 39
	Unaudited Quarterly Financial Information	50
		10-K Pages
(2)	Financial Statement Schedule:	
	Report of Independent Accountants on Financial Statement Schedule for the three years ended December 31, 2000	13
	Valuation and Qualifying Accounts and Reserves (Schedule II)	14 - 16
	Supplementary Information — Commitments and Contingencies	17
	All other schedules are omitted because they are not applicable or the required information is shown in the financial statements or notes thereto.	
(3)	Exhibits listed in the "Exhibit Index"	18 - 19
	Exhibits Nos. 10-A through 10-K are management contracts or compensatory plans or arrangements required to be filed pursuant to Item 14(c) of this report.	
(b)	Reports on Form 8-K	
	None	

Report of Independent Accountants on
Financial Statement Schedule

To the Board of Directors
of Dana Corporation

Our audits of the consolidated financial statements referred to in our report dated February 2, 2001 appearing in the 2000 Annual Report to Shareholders of Dana Corporation (which report and consolidated financial statements are incorporated by reference in this Annual Report on Form 10-K) also included an audit of the financial statement schedule listed in Item 14(a)(2) of this Form 10-K. In our opinion, this financial statement schedule presents fairly, in all material respects, the information set forth therein when read in conjunction with the related consolidated financial statements.

PricewaterhouseCoopers LLP

Toledo, Ohio
February 2, 2001

DANA CORPORATION AND CONSOLIDATED SUBSIDIARIES

SCHEDULE II(a) — VALUATION AND QUALIFYING ACCOUNTS AND RESERVES

ALLOWANCE FOR DOUBTFUL ACCOUNTS RECEIVABLE

	Balance at beginning of period	Additions charged to income	Trade accounts receivable "written off" net of recoveries	Adjustments arising from change in currency exchange rates and other items	Balance at end of period
Year ended –					
December 31, 1998	\$33,942,000	\$20,694,000	\$(16,698,000)	\$2,516,000	\$40,454,000
December 31, 1999	40,454,000	15,521,000	(11,407,000)	(752,000)	43,816,000
December 31, 2000	43,816,000	26,467,000	(28,893,000)	580,000	41,970,000

DANA CORPORATION AND CONSOLIDATED SUBSIDIARIES

SCHEDULE II(b) — VALUATION AND QUALIFYING ACCOUNTS AND RESERVES

ALLOWANCE FOR CREDIT LOSSES — LEASE FINANCING

	Balance at beginning of period	Additions charged to income	Amounts "written off" net of recoveries	Adjustments arising from change in currency exchange rates and other items (1)	Balance at end of period
Year ended –					
December 31, 1998	\$52,653,000	\$20,117,000	\$(10,561,000)	\$(29,537,000)	\$32,672,000
December 31, 1999	32,672,000	8,172,000	(6,000)	—	40,838,000
December 31, 2000	40,838,000	8,421,000	(6,548,000)	(15,000)	42,696,000

(1) Other items in 1998 include \$(28,889,000) from the sale of the Technology Leasing Group portfolio.

DANA CORPORATION AND CONSOLIDATED SUBSIDIARIES

SCHEDULE II(c) — VALUATION AND QUALIFYING ACCOUNTS AND RESERVES

VALUATION ALLOWANCE FOR DEFERRED TAX ASSETS

	Balance at beginning of period	Additions charged to income	Amounts "written off" net of recoveries	Adjustments arising from change in currency exchange rates and other items	Balance at end of period
Year ended –					
December 31, 1998	\$30,400,000	\$28,800,000	—	—	\$ 59,200,000
December 31, 1999	59,200,000	24,000,000	—	—	83,200,000
December 31, 2000	83,200,000	24,000,000	\$(5,100,000)	—	102,100,000

DANA CORPORATION AND CONSOLIDATED SUBSIDIARIES

SUPPLEMENTARY INFORMATION TO FINANCIAL STATEMENTS

COMMITMENTS AND CONTINGENCIES

We are a party to various legal proceedings (judicial and administrative) arising in the normal course of business, including proceedings which involve environmental and product liability claims. You can find additional information in “Note 17. Commitments and Contingencies” on page 37 of our 2000 Annual Report.

With respect to environmental claims, we are involved in investigative and/or remedial efforts at a number of locations, including “on-site” activities at currently or formerly owned facilities and “off-site” activities at “Superfund” sites where we have been named as a potentially responsible party. You can find more information in “Environmental Compliance and Remediation” under “Note 1. Summary of Significant Accounting Policies” on page 27 and “Management’s Discussion and Analysis of Financial Condition and Results of Operations” on pages 40-46 of our 2000 Annual Report.

With respect to product liability claims, we are named in proceedings involving alleged defects in our products. Such proceedings currently include a large number of claims (most of which are for relatively small damage amounts) based on alleged asbestos-related personal injuries. At December 31, 2000, approximately 70,000 such claims were outstanding, of which approximately 33,000 were settled pending payment. We have agreements with our insurance carriers providing for the payment of a significant majority of the indemnity costs and the legal and administrative expenses for these claims. You can find additional information under “Management’s Discussion and Analysis of Financial Condition and Results of Operations” on pages 40-46 of our 2000 Annual Report.

EXHIBIT INDEX

No.	Description	Method of Filing
3-A	Restated Articles of Incorporation	Filed by reference to Exhibit 3-A to our Form 10-Q for the quarter ended June 30, 1998
3-B	By-Laws, effective October 16, 2000	Filed by reference to Exhibit 3-B to our Form 10-Q for the quarter ended September 30, 2000
4-A	Specimen Single Denomination Stock Certificate	Filed by reference to Exhibit 4-B to our Registration Statement No. 333-18403 filed December 20, 1996
4-B	Rights Agreement, dated as of April 25, 1996, between Dana and ChemicalMellon Shareholder Services, L.L.C., Rights Agent	Filed by reference to Exhibit 1 to our Form 8-A filed May 1, 1996
4-C	Indenture for Senior Securities between Dana and Citibank, N.A., Trustee, dated as of December 15, 1997	Filed by reference to Exhibit 4-B of our Registration Statement No. 333-42239 filed December 15, 1997
4-D	First Supplemental Indenture between Dana, as Issuer, and Citibank, N.A., Trustee, dated as of March 11, 1998	Filed by reference to Exhibit 4-B-1 to our Report on Form 8-K dated March 12, 1998
4-E	Form of 6.5% Notes due March 15, 2008 and 7.00% Notes due March 15, 2028	Filed by reference to Exhibit 4-C-1 to our Report on Form 8-K dated March 12, 1998
4-F	Second Supplemental Indenture between Dana, as Issuer, and Citibank, N.A., Trustee, dated as of February 26, 1999	Filed by reference to Exhibit 4.B.1 to our Form 8-K dated March 2, 1999
4-G	Form of 6.25% Notes due 2004, 6.5% Notes due 2009, and 7.0% Notes due 2029	Filed by reference to Exhibit 4.C.1 to our Form 8-K dated March 2, 1999
10-A	Additional Compensation Plan	Filed by reference to Exhibit A to our Proxy Statement dated March 3, 2000
10-B	1997 Stock Option Plan	Filed by reference to Exhibit A to our Proxy Statement dated March 5, 1999
10-B(1)	First Amendment to 1997 Stock Option Plan	Filed by reference to Exhibit B to our Proxy Statement dated March 2, 2001
10-C	Excess Benefits Plan	Filed by reference to Exhibit 10-F to our Form 10-K for the year ended December 31, 1998

No.	Description	Method of Filing
10-C(1)	First Amendment to Excess Benefits Plan	Filed by reference to Exhibit 10-C(1) to our Form 10-Q for the quarter ended September 30, 2000
10-D	Director Deferred Fee Plan	Filed by reference to Exhibit B to our Proxy Statement dated February 28, 1997
10-D(1)	First Amendment to Director Deferred Fee Plan	Filed by reference to Exhibit 10-I(1) to our Form 10-Q for the quarter ended March 31, 1998
10-D(2)	Second Amendment to Director Deferred Fee Plan	Filed by reference to Exhibit 10-I(2) to our Form 10-K for the year ended December 31, 1998
10-E	Employment Agreement between Dana and J.M. Magliochetti	Filed with this Report
10-F	Change of Control Agreement between Dana and W.J. Carroll. There are substantially similar agreements with R.L. Clayton, B.N. Cole, M.A. Franklin, C.F. Heine, J.M. Laisure, R.C. Richter, and E.J. Shultz	Filed by reference to Exhibit 10-J(4) to our Form 10-K for the year ended December 31, 1997
10-G	Collateral Assignment Split-Dollar Insurance Agreement for Universal Life Policies between Dana and J.M. Magliochetti. There are substantially similar agreements with W.J. Carroll, M.A. Franklin, and E.J. Shultz.	Filed with this Report
10-H	Supplemental Benefits Plan	Filed by reference to Exhibit 10-K to our Form 10-K for the year ended December 31, 1998
10-I	1999 Restricted Stock Plan	Filed by reference to Exhibit B to our Proxy Statement dated March 5, 1999
10-J	1998 Directors' Stock Option Plan	Filed by reference to Exhibit A to our Proxy Statement dated February 27, 1998
10-K	Supplementary Bonus Plan	Filed by reference to Exhibit 10-N to our Form 10-Q for the quarter ended June 30, 1995
13	Those sections of our 2000 Annual Report that are referred to in this Form 10-K	Filed with this Report
21	Subsidiaries of Dana	Filed with this Report
23	Consent of PricewaterhouseCoopers LLP	Filed with this Report
24	Power of Attorney	Filed with this Report

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

		DANA CORPORATION
		(Registrant)
Date:	March 1, 2001	By: /s/ Michael L. DeBacker
		Michael L. DeBacker, Vice President

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the date indicated.

Date:	March 1, 2001	/s/ Joseph M. Magliochetti
		Joseph M. Magliochetti, Chairman of the Board
Date:	March 1, 2001	/s/ Robert C. Richter
		Robert C. Richter, Chief Financial Officer
Date:	March 1, 2001	/s/ Charles W. Hinde
		Charles W. Hinde, Chief Accounting Officer
Date:	March 1, 2001	*/s/ B.F. Bailar
		B.F. Bailar, Director
Date:	March 1, 2001	*/s/ A.C. Baillie
		A.C. Baillie, Director
Date:	March 1, 2001	*/s/ E.M. Carpenter
		E.M. Carpenter, Director
Date:	March 1, 2001	*/s/ E. Clark
		E. Clark, Director
Date:	March 1, 2001	*/s/ G.H. Hiner
		G.H. Hiner, Director
Date:	March 1, 2001	*/s/ M.R. Marks
		M.R. Marks, Director
Date:	March 1, 2001	*/s/ R.B. Priory
		R.B. Priory, Director

Purchase Orders: Purchase orders are effective and binding upon the contractor when placed in the mail.

Quantities: The quantities listed are estimated only and based upon the filed requirements. However, vendor payments will be based on quantities actually ordered and received during the contract period.

Minimum Order: No minimum order.

Contract Period: It is the intention of the State to enter into a contract for a term of (initial contract term), as indicated on the IFBs. Agency reserves the right to extend this contract under the same terms and conditions for an additional period of (allowed in IFB but not to exceed a total contract term of 5 years).

Price Adjustment: The contractor may request a price change for any extension period(s) if allowed and limited to the price adjustment clause(s) per the IFB. Documentation of the increases must accompany the request. Price reductions may be offered at any time during the term of the contract including renewal periods.

Appendix A: The contractor agrees to comply with Appendix A, New York State Standard Contract Clauses.

Terms and Conditions: The contractor agrees to comply with all the specifications of (agency) Bid Number (x) dated (specific date).

Notes to Agency as applicable:

- Purchase orders should reference this contract number (PAxx)
- When placing orders against this contract, please forward a copy of the P.O. to: (Agency main office, address, phone number, fax number)
- Agencies are urged to process vouchers expeditiously.

(see item list on next page)

CONTRACT NO. PAXx - Public Safety Officers Uniforms (SAMPLE ONLY)

Item Number 1:

Estimated Quantity: 1005

Name and number of material proposed to be furnished: Raeford 571-34453

Style number or catalog reference of finished garment: Special

Manufacturer: Clifton Shirt

Delivery: 45 days A/R/O (after start up)

Item: Long Sleeve Shirts (men)
with embroidered insignia patch, as per specification
Color: Blue/Gray

Sizes:

Neck 14 C 17 2, Sleeve 31-35 **Unit Price:** 42.70

Neck 18 C 20 2, Sleeve 36-38, or long body **Unit Price:** 46.70

Item Number 2:

Estimated Quantity: 125

Name and number of material proposed to be furnished: Raeford 571-34453

Style number or catalogue reference of finished garment: Special

Manufacturer: Clifton Shirt

Delivery: 45 days A/R/O (after start up)

Item: Long Sleeve Shirts (women)
with embroidered insignia patch, as per specification
Color: Blue/Gray

Sizes:

S,M,L **Unit Price:** 42.70

XL **Unit Price:** 46.70

List of Approved Key Personnel for N66604-26-R-0163

The following Key Personnel are approved for this task order. Any changes to the below list require approval in accordance with clause C-237-H002 "SUBSTITUTION OF KEY PERSONNEL".

[illegible]

LOK SABHA

A

BILL

to provide for the readjustment of the allocation of seats in the House of the People to the States and Union territories, the total number of seats in the Legislative Assembly of each State and Union territories having a Legislative Assembly, the division of each State and each Union territory having a Legislative Assembly into territorial constituencies for elections to the House of the People and Legislative Assemblies of the States and Union territories and for matters connected therewith.

*(Shri Arjun Ram Meghwal, Minister of State (I/C) for Law & Justice and
Minister of State of Parliamentary Affairs)*

increases as the focus narrows to small firms only, suggesting that smaller enterprises bear a disproportionate cost of a weak business environment. For example, being credit constrained is associated with a reduction in productivity of -1.64 log points for the average firm but -4.39 log points for small firms. By contrast, access to a bank account is strongly and positively associated with productivity, again with effects that grow larger as firm size decreases.

Third, the sign and magnitudes of these effects are economically meaningful and broadly consistent with findings in the existing literature. The strong negative effect of corruption is in line with Fisman and Svensson (2007) and Sequeira and Djankov (2014), who document substantial productivity losses from corruption in African contexts. The disproportionate burden of tax-related constraints on smaller firms resonates with Aterido, Hallward-Driemeier, and Pagés (2011), who show that regulatory and tax harassment disproportionately constrain small enterprise growth. Likewise, the importance of financial access echoes the evidence from Beck, Demirgüç-Kunt, and Maksimovic (2005), who find that credit constraints significantly hinder smaller firms' performance, while Ayyagari, Demirgüç-Kunt, and Maksimovic (2011) emphasize the critical role of financial inclusion for productivity growth in SMEs. Finally, the overall pattern of stronger negative effects for smaller firms is consistent with the broader literature on firm heterogeneity, which argues that weak institutional and financial environments amplify vulnerabilities among smaller and less established enterprises.

constraint	All firms	Small and Medium firms	Small firms only
regulation	-0.0220***	-0.0211***	-0.0374***
tax time	-0.0367	-0.0414	-0.126**
tax pressure	-0.210	-0.257	-0.477*
corruption	-0.519	-0.905	-2.053***
corruption intensity	-0.0863***	-0.0838***	-0.202***
bank account	1.153**	1.892***	3.227***
credit constrained	-1.636**	-1.653**	-4.389***

Table 4: Effect of business constraints on firm-level productivity

To facilitate the interpretation of magnitudes, we conduct a counterfactual exercise based on regressions that include all constraints simultaneously. We then assess the productivity gains that would result from halving the level of each constraint—bringing them roughly back to the level observed among peer countries. The results, reported in Figure 20, highlight several striking patterns. First, the estimated productivity gains are economically significant across most constraints, but especially large among small firms. For example, reducing corruption intensity by half would raise productivity by 50 percent for small firms, compared to 19 percent for the average firm. Similarly, alleviating tax-related burdens yields relatively modest productivity gains at the aggregate level (around 11–12 percent), but substantial improvements for small firms, with tax time reductions associated with productivity gains of up to 46 percent.

Second, financial inclusion emerges as a particularly powerful channel: improving access to bank accounts is associated with a 45 percent increase in productivity for the average firm and more than a doubling (109 percent) among small firms. Access to a bank account is a critical enabler of firm growth because it allows businesses to securely store funds, make and receive payments efficiently, and establish a financial history that is vital for accessing credit. In the DRC context, this constraint is particularly binding due to heavy reliance on cash, limited banking infrastructure outside major urban centers, and extensive documentation requirements (e.g., business registration, tax ID) that most small firms and informal entrepreneurs struggle

to meet. Moreover, since most banks require an account relationship as a prerequisite for loan applications, lack of access to an account effectively locks firms out of the broader financial ecosystem. This helps explain the unusually high productivity impact associated with improved access to bank accounts, particularly for small firms. Finally, credit constraints remain a major impediment to productivity, with small firms again disproportionately affected. These simulated magnitudes underscore the economic relevance of the regression estimates and provide a strong case for policy interventions targeted at corruption reduction, regulatory simplification, and financial deepening, particularly with a focus on small enterprises.

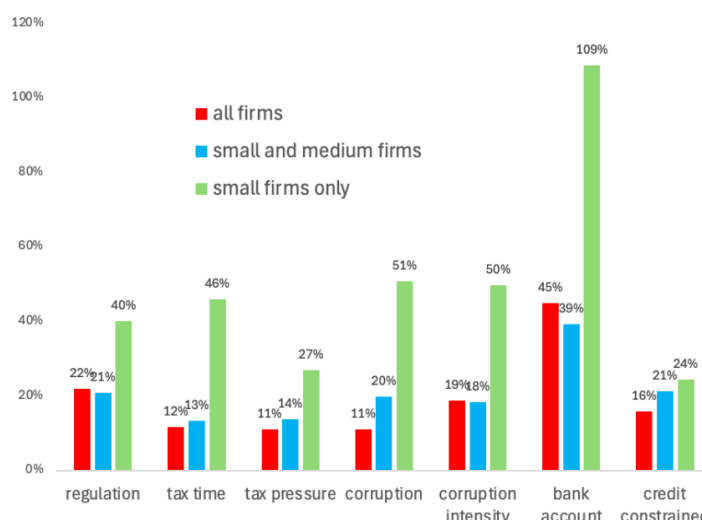


Figure 20: Boost to firm-level productivity after reduction in constraints

It should be noted that the counterfactual impacts documented in Figure 20 are based on IV estimates intended to mitigate reverse causality and omitted-variable bias. Nonetheless, because the instruments are constructed from local (city $\times$ size) averages, they may reflect broader local conditions that affect firm performance through multiple channels; as a result, the estimated impacts may overstate the purely firm-level causal effect of the targeted constraint and are best interpreted as (i) capturing the consequences of improving the local business environment along that margin (ii) an upper bound on the firm-level causal impact of a given constraint. Separately, baseline productivity among small firms is extremely low in the data, so percentage changes can appear large even when the implied absolute gains are very modest. For this reason, proportional increases of 30–50 percent for the smallest firms are not necessarily implausible, but the estimates should be interpreted as indicative orders of magnitude rather than precise point predictions

Finally, Table 5 summarizes the results from estimating equation (2) with investment, R&D spending, and innovation as dependent variables. Three key findings emerge. First, access to finance is unsurprisingly a strong predictor of firm investment. The coefficient on borrowing is large and highly significant across all specifications (2.117 for all firms, 2.404 for SMEs, and 3.293 for small firms), suggesting that access to credit more than doubles the probability of investment, and even more so for small firms. This aligns with evidence that easing financial constraints is critical to stimulate capital accumulation and growth in developing countries (Beck et al., 2008; Ayyagari et al., 2011). It further underscores the importance of ongoing reforms to strengthen credit infrastructure in the DRC—such as operationalizing a modern credit reporting system and creating a centralized electronic collateral registry for movable assets—to reduce information asymmetries and relax collateral constraints. Coupled with the rollout of partial credit guarantees under the PPCG Fund, these measures could lower perceived risk and unlock financing for MSMEs. These efforts are fully aligned with the Stratégie Nationale d’Inclusion Financière (SNIF) 2023–2028, which prioritizes expanding access to financial services and increasing credit to MSMEs through systemic improvements in credit infrastructure and risk-sharing mechanisms.

Second, corruption appears to facilitate investment and R&D spending. Both the likelihood of facing corruption and the intensity of corruption are positively and significantly associated with investment and R&D, particularly among SMEs. These findings are consistent with the “grease the wheels” hypothesis, whereby informal payments help firms bypass bureaucratic frictions and regulatory bottlenecks (Fisman & Svensson, 2007). However, the longer-run literature warns that reliance on bribery undermines competition and resource allocation efficiency. Finally, competition from the informal sector significantly reduces the likelihood of innovation. The negative coefficients are strong and significant across all specifications (–1.909 for all firms, –1.961 for SMEs, and –3.825 for small firms). This suggests that informal competition also discourages innovation, in line with evidence that informality distorts market incentives and reduces innovation capacity.

	all firms	small and medium firms	small firms only
Investment decision			
corruption	1.753**	1.674**	1.923**
corruption intensity	0.0999***	0.105***	0.230***
borrowing	2.117***	2.404***	3.293***
R&D Spending			
corruption	2.455***	2.010**	1.171
corruption intensity	0.0978***	0.101***	0.0795
Innovation			
informal	-1.909**	-1.961*	-3.825**

Table 5: effect of several business constraints on firm-level investment, R&D and Innovation

VI. Conclusion

This paper has provided the first comprehensive analysis of the 2024 World Bank Enterprise Survey for the Democratic Republic of Congo, offering new evidence on the structure of the private sector, the incidence of business environment constraints, and their effects on firm performance. The findings paint a sobering picture: despite the country’s resource wealth and large market potential, the private sector remains dominated by micro and small firms operating in low-productivity activities, with limited structural transformation since 2013. The persistence of widespread regulatory burdens, excessive taxation, corruption, and financing constraints continues to undermine the capacity of firms to scale up, innovate, and contribute to growth.

A key insight of this study is that these barriers do not fall evenly across firms but disproportionately affect the very enterprises that could drive economic transformation—smaller and younger businesses, exporters, and high-performing firms. This asymmetry reduces allocative efficiency and suppresses the emergence of competitive and innovative firms, while protecting low-performing incumbents. Econometric estimates confirm that the cumulative effect of these constraints imposes a substantial drag on productivity, with simulations suggesting that easing them to peer-country levels could raise output per and encourage investment and innovation. More broadly, this paper contributes to the literature on firm productivity in developing and fragile states by demonstrating how country-specific enterprise surveys can inform both research and policy dialogue. The DRC’s experience illustrates how a hostile business environment not only constrains aggregate productivity but also distorts the growth trajectory by filtering out the most

dynamic segments of the private sector. Addressing these distortions is essential if the country is to unleash its entrepreneurial potential and lay the foundations for sustained and inclusive growth.

From a policy perspective, the evidence underscores the urgency of business environment reforms in the DRC. Priority areas include reducing regulatory harassment, addressing corruption, rationalizing the tax system, and, most critically, expanding access to finance for small enterprises. Given the substantial regional disparities highlighted in the data, reform strategies should also account for subnational variation and leverage regions with high economic potential.

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